US-China Decoupling and Board Composition: Evidence from Chinese Listed Firms

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Abstract

When the United States imposed sanctions and investment bans on Chinese firms, U.S.-connected directors of Chinese listed companies became personally exposed to legal and compliance risk under the U.S. Persons Rule. Because the timing and targeting of this shock were set by U.S. national-security policy and were largely external to firm governance, we treat it as quasi-exogenous and ask how Chinese boards adjusted. Since standard corporate data do not report director nationality or U.S.-nexus status, we construct a classification pipeline that anchors an auditable three-prong U.S.-nexus definition in structured CSMAR records and supplements unresolved cases with web-scraped biographies labeled by a large-language model, applying the same measurement architecture across the mainland A-share and Hong Kong markets. We find that exposed mainland firms shed roughly two percentage points of U.S.-nexus director share, with the effect concentrated among independent directors and realized through non-renewal at term end. The exit appears even at firms under no firm-specific prohibition, which we interpret as compliance chill operating through individual liability. Substitution is market-specific: mainland boards recruit domestically credentialed professionals, while Hong Kong boards recruit non-U.S. internationals. The exit itself remains selective, falling on U.S.-connected directors, with no comparable departure among foreign directors more generally.

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1 Introduction

In recent years, prominent Chinese listed firms have watched their U.S.-connected directors leave their boards as Washington and Beijing have pulled their economies apart. In March 2023, for instance, two naturalized U.S. citizens, Du Zhiyou and Huang Qing, resigned from the board of Advanced Micro-Fabrication Equipment (AMEC), a Shanghai-listed maker of chip-manufacturing equipment, citing personal reasons only months after Washington had barred "U.S. persons" from supporting advanced chipmaking in China.[1] The firm's own founder and chairman, a Silicon Valley veteran of Intel and Applied Materials, would himself renounce his U.S. citizenship and reclaim Chinese nationality.[2] A co-founder of Arm (a British chip designer), who resigned from the board of Semiconductor Manufacturing International Corporation (SMIC) after nine years, observed that "the international divide has further widened."[3] Over this period, the United States and China have undergone a marked process of economic and financial decoupling, with a growing literature documenting its consequences for trade, innovation, and firm value (Egger and Zhu, 2020; He et al., 2022).

A fundamental question nonetheless remains unanswered: through what internal governance mechanisms do geopolitical shocks reshape firms? This Article addresses that question by examining how U.S.–China decoupling has reshaped board composition in Chinese listed firms, and by asking who replaces the directors who depart. We document board reconstitution as a previously underexamined margin along which geopolitical fragmentation reaches the firm, a margin that operates through the market for director human capital and that is distinct from the trade and financial channels emphasized in prior work.

Our analysis proceeds in two steps. First, we ask whether U.S. sanctions reshape board composition, that is, whether they reduce the share of directors with U.S. backgrounds on the boards of affected Chinese firms, and, when a seat is vacated, whether it is filled by a domestic professional, by a state-affiliated figure, or left empty. We then ask whether the same shock reshaped boards in an independent cross-market setting, namely Hong Kong-listed Chinese firms, where the foreign investor base is directly measurable and its response can be observed.

Answering these questions requires a variable that standard corporate-governance data do not report—the nationality and U.S.-nexus of individual directors—which must be constructed from director-level records. We anchor an auditable three-prong U.S.-nexus definition (i.e., U.S. nationality or permanent residency, substantial U.S. work experience, or U.S. professional licensure) in structured records from the China Stock Market and Accounting Research (CSMAR) database,

[1] "Tech war: US directors resign from board of Shanghai-listed chip equipment firm amid Washington restrictions on citizens," *South China Morning Post*, 15 March 2023, https://www.scmp.com/tech/tech-war/article/3213626/tech-war-us-directors-resign-board-shanghai-listed-chip-equipment-firm-amid-washington-restrictions.

[2] "Tech war: founder of Chinese semiconductor equipment maker AMEC renounces US citizenship," *South China Morning Post*, 18 April 2025, https://www.scmp.com/tech/tech-war/article/3307075/tech-war-founder-chinese-semiconductor-equipment-maker-amec-renounces-us-citizenship.

[3] "Ex-Arm boss exits Chinese chipmaker amid US-China tensions," *Al Jazeera*, 11 August 2022, https://www.aljazeera.com/economy/2022/8/11/ex-arm-boss-exits-chinese-chipmaker-amid-us-china-tensions; "Tudor Brown resigns from SMIC," *Electronics Weekly*, 11 August 2022, https://www.electronicsweekly.com/news/business/tudor-brown-resigns-from-smic-2022-08/.

and for directors whose connections are not verifiable from structured fields we supplement it with web-scraped biographical text classified by a large-language model, cross-checking it against an independent nationality classifier and applying it consistently across the mainland and Hong Kong markets. We treat this measurement as the methodological foundation for the analysis, with its full construction and validation reported in Appendix A.1.

We find that U.S.–China decoupling reduces the proportion of directors with U.S. backgrounds on the boards of Chinese listed firms by approximately two percentage points, roughly one-third of the pre-shock treated-firm mean. We further find that these directors are replaced predominantly by domestically credentialed professionals, and that departure occurs through passive non-renewal at term end, with forced mid-term resignation playing little role. Because the design is best suited to recovering the exit and leaves the subsequent staffing decision to descriptive analysis, we interpret the exit as causal—subject to the threats to identification we address in Section 7—while treating the replacement pattern as a descriptive characterization of board recomposition.

The Article makes two contributions. The first concerns the individual-liability mechanism behind the exit. The exit appears across the full set of firms with a U.S.-connected director, including firms under no direct legal prohibition, a breadth we read as compliance chill: heightened personal legal and reputational risk under the U.S. Persons Rule plausibly raised the cost of board service for U.S.-connected directors even where no firm-specific ban applied. A within-firm placebo separates this U.S.-specific channel from a concurrent market-wide retreat of foreign directors from Chinese boards, because at exposed firms only U.S.-connected directors leave, non-U.S. foreign directors are unaffected on the mainland and actively recruited in Hong Kong, and firms with non-U.S. foreign exposure but no U.S. connection lose those directors regardless of sanctions. To our knowledge this is the first paper to isolate the sanctions-specific component of board de-internationalization from the broader Sino-Western governance fragmentation trend. Because these departing directors were disproportionately independent monitors, we frame the loss as the governance consequence that gives the mechanism its stakes and test it directly in Section 6, where firm-level crash risk, the market's reaction to the departures, executive pay-performance sensitivity, and payout policy all return bounded nulls (i.e., no measurable near-term governance deterioration), consistent with the one-for-one domestic substitution we document. Although a decoupling literature has documented broad stock-market reactions and spillovers from U.S.–China tensions (Egger and Zhu, 2020; He et al., 2022; Oh and Kim, 2024) and has shown that sanctions and export controls reshape cross-border investment and strategic adjustment (Biglaiser and Lektzian, 2011; Hu et al., 2024; Liu and Li, 2023), it has largely left unopened the internal channel through which those effects reach the firm; by tracing the exit and the recomposition, we identify board composition as one such channel. A complementary strand examines a different internal channel at the same governance–geopolitics intersection, showing that U.S. asset managers' proxy voting on Chinese charter amendments that formalize Chinese Communist Party committees reflects a tension between their commercial interests and broader U.S. policy objectives (Lin, 2026; Lin and Milhaupt, 2021), whereas the channel we document operates through the market for director human capital rather than through investor voting.

The second contribution is cross-market replication. Extending the analysis to Hong Kong-listed

Chinese firms, where the foreign investor base is directly measurable, reproduces the director-exit result in a distinct regulatory and ownership environment and shows that substitution is market-specific: mainland boards draw on a domestic talent pool, whereas Hong Kong boards draw on a deep non-U.S. international pool. Financial ownership appears to respond more narrowly than board composition, with measurable reallocation surfacing only among the explicitly sanctioned Hong Kong firms; we report that capital-base evidence in Appendix A.2 and read it cautiously, since the mainland and intent-to-treat ownership nulls partly reflect low baseline foreign exposure and therefore bear weakly on investor behavior.

These questions also carry policy implications. By documenting the exit of U.S.-connected directors and the domestic professionals who replaced them, this Article helps policymakers assess how geopolitical fragmentation reshapes corporate boards and how deep the domestic director-labor market is that absorbs such shocks. The pattern also points to an unintended consequence of extraterritorial sanctions. Because the directors who departed were disproportionately independent directors with international credentials—the type that a substantial literature associates with stronger monitoring and with constraining controlling-shareholder expropriation in Chinese firms (Zhu et al., 2016; Gong et al., 2021; Jiang et al., 2010)—a measure aimed at restricting U.S. capital flows may have incidentally removed a potential source of minority-shareholder protection from the affected boards. Whether that removal imposed an actual cost on domestic investors remains an open question: our consequence analysis in Section 6 detects no measurable near-term deterioration in crash risk, in the market's reaction to the departures, or in executive pay-performance sensitivity and payout policy, a bounded null we read alongside the one-for-one domestic substitution, and longer-horizon evidence would be needed to settle the point within the broader policy conflict between the sanctioning and sanctioned governments.

The remainder of the Article proceeds as follows. Section 2 describes the U.S. sanctions framework and the individual-liability channel and reviews the related literature. Section 3 sets out the data, the measurement of U.S.-nexus board connections, and the empirical strategy, whose causal weight rests on within-firm and within-board comparisons. Section 4 reports the exit of U.S.-nexus directors, the board reconstitution that follows, and the placebo decomposition that isolates the U.S.-specific channel. Section 5 presents the cross-market replication in Hong Kong. Section 6 examines whether the loss of these directors matters for firm governance and reports a bounded null. Section 7 discusses limitations and extensions.

2 Institutional Background and Related Literature

2.1 Sanctions and Individual Liability

Beginning in 2019, the United States enacted a series of executive orders that imposed sanctions on Chinese listed firms. President Trump first signed Executive Order 13873, which authorized restrictions on transactions involving Chinese firms operating in the information and communications technology and services sector on national-security grounds. In 2020, the administration issued Executive Order 13959, which for the first time explicitly prohibited U.S. investors from investing in firms identified as "Communist Chinese military companies," that is, companies deemed to have ties to the Chinese military.

The Biden administration largely preserved and extended this policy direction. In 2021, it issued Executive Order 14032, which retained the earlier investment restrictions while broadening their scope. Building on these executive orders, the Office of Foreign Assets Control (OFAC) systematically established and updated the NS-CMIC list (i.e., the Non-SDN Chinese Military-Industrial Complex Companies List). According to official announcements, the first large-scale inclusion of Chinese firms under U.S. investment sanctions took place on November 12, 2020. OFAC subsequently undertook a major revision and expansion of the list on June 3, 2021, and the unified effective date for many of the investment prohibitions was set as August 2, 2021, from which point U.S. investors were formally prohibited from making new securities investments and capital allocations to the affected firms.

These prohibitions carry direct implications for directors who qualify as U.S. persons and serve on the boards of affected Chinese listed companies. The threshold concept is the definition of a "U.S. person," which the sanctions regime construes broadly to reach any U.S. citizen, lawful permanent resident, entity organized under U.S. law (including its foreign branches), and any individual physically present in the United States, irrespective of where the person resides or where the conduct occurs. A director who falls within this definition is bound by the prohibitions personally, independent of the nationality of the firm on whose board the director sits.

The operative prohibition, codified in the Chinese Military-Industrial Complex Sanctions Regulations (31 C.F.R. part 586), bars U.S. persons from the purchase or sale of any publicly traded securities of a listed NS-CMIC firm, together with derivatives of, and instruments designed to provide investment exposure to, such securities. Although board service as such is not enumerated among the prohibited acts, the individual-liability exposure that concerns us arises at the intersection of three features of the regime. First, the regulations separately prohibit any transaction that evades or avoids, has the purpose of evading or avoiding, or causes a violation of the prohibitions, together with conspiracies and attempts to do so, so that a U.S.-person director who approves or authorizes a covered securities transaction by the firm, such as an issuance, a buyback, or treasury dealing in covered instruments, may thereby cause or facilitate a violation even without trading on personal account. Second, equity-linked director compensation, including shares, options, or restricted units in a covered firm, can itself constitute the acquisition of a prohibited security. Third, because civil liability under the International Emergency Economic Powers Act attaches on a strict-liability basis, without proof of willfulness, a director's good-faith uncertainty about whether a particular board decision touches a covered security does not avert exposure.

The associated penalties are severe enough to make this exposure salient even where the probability of enforcement is modest. The International Emergency Economic Powers Act authorizes civil penalties up to the greater of a statutory maximum or twice the value of the offending transaction and, for willful violations, criminal penalties of up to $1,000,000 and twenty years' imprisonment.[4]

[4] 50 U.S.C. § 1705; the prohibitions and the "U.S. person" definition are codified at 31 C.F.R. part 586. The statutory civil maximum is adjusted periodically for inflation, and the precise regulatory cross-references should be confirmed against the OFAC regulations current at the time of submission.

The regime did, however, grant U.S. persons time-limited general licenses to wind down and divest covered positions within specified windows, which concentrated the practical pressure to exit covered relationships around the 2021 effective dates and avoided a diffuse temporal spread.

This role-, compensation-, and transaction-contingent exposure is the source of the individual-level legal and reputational risk that we later term compliance chill. The exposure is most acute for independent directors, who typically receive the fee- or equity-based compensation that falls within the regulations' reach and who, because they lack an executive or controlling stake, bear lower switching costs than insider directors once the legal cost of continued service rises.

Beyond direct legal exposure, these executive orders, together with the broader sanctions framework directed at Chinese listed companies, plausibly created substantial constraints for U.S.-national directors even where no explicit prohibition applied. In practice, such directors may have found it increasingly difficult, and in some cases impractical, to continue serving on the boards of sanctioned or politically sensitive Chinese listed firms as legal, compliance, and reputational risks rose. The increasingly hostile geopolitical environment may also have encouraged foreign directors, as well as internationally connected directors based in China, to resign from Chinese boards even absent a direct legal prohibition, since the reputational and regulatory costs of such associations grew harder to justify. The mechanism described above connects to two strands of prior work, one on the geopolitics of board composition and one on foreign-director human capital, which we review in turn.

2.2 U.S.–China Decoupling and Board Composition: Exit, Replacement, and Domestic Professionalization

Recent evidence suggests that director retention is sensitive to the geopolitical environment, because board service is embedded in cross-border legal, reputational, and identity contexts. As Tian et al. (2025) document, foreign directors are more likely to exit Chinese boards when bilateral political relations deteriorate. Complementing this identity-based mechanism, Liu et al. (2025) show that sanctions can generate "shock anticipation" in director labor markets, so that foreign directors may resign when their own firms are directly targeted and when sanctions imposed on peer firms raise perceived legal and reputational risk. This anticipatory resignation logic is consistent with the broader finding that outside directors tend to quit preemptively when they foresee reputational or legal exposure ahead (Fahlenbrach et al., 2010).

A board human capital perspective, however, requires that we move beyond exit alone (Adams et al., 2010). Because board seats are typically refilled, the identity and expertise of replacement directors can meaningfully change a board's human capital and governance capacity. Indeed, a growing literature on China suggests that specific kinds of professional directors matter for governance. Zhu et al. (2016) show that more empowered independent directors engage in stronger monitoring and are associated with higher firm value, while Chen et al. (2019) show that forced resignations of academic independent directors in China reduce firm value, which suggests that academically trained directors contribute economically meaningful expertise. As He and Liu (2016) document, Chinese listed firms appoint independent directors with professional legal backgrounds for specific consulting and governance purposes, thereby treating independent directors as heterogeneous sources of expertise. More broadly, Firth et al. (2016) show that regulatory sanctions imposed on independent directors in China have persistent consequences for their future board opportunities, which reinforces the plausibility of systematic director-labor-market adjustment after external shocks.

Regulation can also change board composition directly by determining which types of directors may serve. Fan (2021) studies China's regulation banning politicians from corporate boards and shows that this external regulatory shock led to substantial board reconfiguration and affected firm performance. Berkowitz et al. (2022) extend this logic, showing that firms that lost politically connected directors initially suffered a valuation decline but subsequently became more productive, transparent, and innovative, a pattern of short-run disruption followed by longer-run governance improvement. Both papers demonstrate that external political and regulatory shocks can reshape the mix of directors on boards, and that the consequences depend critically on who replaces those who depart. Building on these insights, this Article examines how U.S.–China decoupling pressures drive the exit of U.S.-nexus directors from Chinese listed firms, who replaces them, and what that substitution implies for overall board human capital. In particular, we examine whether firms respond by substituting toward domestically credentialed professional directors when U.S.-connected board human capital becomes more costly to maintain.

2.3 Board Composition and Foreign-Director Human Capital

A large literature studies how board composition affects firm governance and firm value (Adams et al., 2010). The board internationalization literature argues that foreign directors affect firm outcomes through both monitoring and advising channels, including the transfer of expertise and the reduction of information frictions (Masulis et al., 2012; Oxelheim and Randøy, 2003; Oxelheim et al., 2013). Related work suggests that foreign independent directors may be especially valuable where domestic legal institutions are weaker, or where the directors themselves come from stronger legal environments (Miletkov et al., 2017). Evidence from China similarly suggests that directors with overseas experience improve firm performance and the information environment (Giannetti et al., 2015), that foreign experience is associated with lower stock-price crash risk (Cao et al., 2019), and that it is associated with different payout policies (Tao et al., 2022). A related external-certification view argues that foreign-connected directors can enhance credibility with global investors and transaction partners, especially where liability-of-origin concerns are salient (Yu et al., 2024). More broadly, the cross-listing literature establishes the general principle that firms with tighter links to developed-market capital and governance standards command a valuation premium, because those links signal commitment to outside investors (Doidge et al., 2004). Because the departure of U.S.-connected directors in our setting is driven by an external political shock, and is therefore separated from endogenous board choices, we can study how boards recompose when this internationally connected human capital becomes costly to retain, thereby distinguishing the exit from firms' ordinary board-selection decisions.

The Chinese governance literature also suggests that director quality matters for monitoring and value. As Liu et al. (2015) show, board independence has a positive overall effect on firm performance in China; Jiang et al. (2016) document that independent directors' reputation concerns shape their voting behavior; and Gong et al. (2021) find that stronger board independence can reduce tunneling by controlling shareholders. The tunneling channel is especially consequential in the Chinese context, where controlling shareholders have historically used intercorporate loans and related-party transactions to siphon resources from listed companies at the expense of minority shareholders (Jiang et al., 2010).

Two features of this prior work motivate our design. Most corporate-governance research treats board structure as the outcome of firm-level economic choices, such as the need for expertise, monitoring, or external resources (Adams et al., 2010), whereas the sanctions episode we study is an external political shock that alters board composition for reasons unrelated to firms' underlying economic fundamentals. Because boards are chosen by firms and directors self-select onto them, board-composition studies routinely face an endogeneity problem that an externally imposed, sanctions-driven departure helps to overcome, and that opening is what this Article exploits in tracing how a geopolitical shock reshapes the internal organization of Chinese listed firms.

Figure 1 consolidates this Article's identification logic and the compliance-chill mechanism in a single diagram. It traces the quasi-exogenous sanctions shock, the individual-liability channel through which it operates, the director exit we interpret as causal, the descriptive board recomposition that follows, and the selective-versus-indiscriminate threshold around which the cross-market evidence is organized.

3 Data and Empirical Strategy

3.1 Measuring U.S.-Nexus Board Connections

The director-level attribute on which the analysis turns—whether a director holds a meaningful connection to the United States—is not reported in standard corporate-governance data, so we construct it here. Our primary measure classifies a director as U.S.-nexus if any of three conditions holds in CSMAR structured records: U.S. nationality or permanent residency, at least five years of U.S. work experience, or a U.S. professional license, applied to the resume, education, professional-title, nationality, and licensure fields of the CSMAR Corporate Governance module. Because this structured measure is fully auditable and relies on no model-based inference, we use it as the regressor underlying all headline estimates. For directors whose connection cannot be verified from structured fields, we supplement it with a classification layer over web-scraped biographies that defines the expanded treatment definitions used in robustness checks; in the Hong Kong market, where structured nationality coverage is thinner, we infer director origin from a multi-dialect Chinese surname classifier validated against an independent nationality classifier. Appendix A.1 details the full measurement pipeline, the classification layer, and the surname validation, and reports that the headline exit estimate is essentially unchanged once classified directors are added.

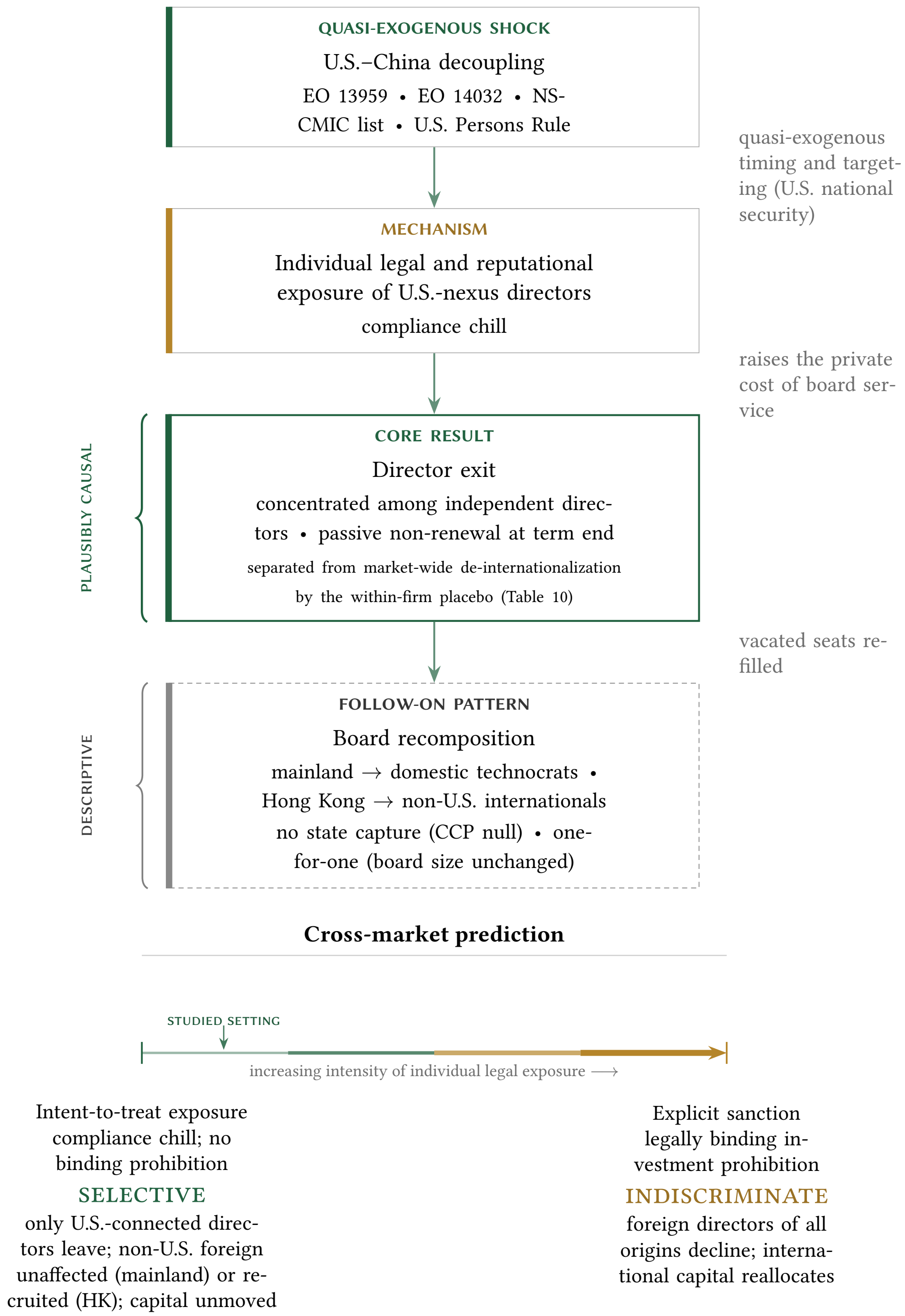

Figure 1: Identification and mechanism. The vertical chain traces the causal architecture: a quasi-exogenous U.S. sanctions shock raises the individual legal and reputational exposure of U.S.-nexus directors (compliance chill), which we interpret as causing the director exit, around which boards descriptively recompose. The within-firm placebo decomposition (Table 10)—under which only U.S.-connected directors leave while non-U.S. foreign directors at the same firms do not—separates the U.S.-specific channel from a concurrent market-wide de-internationalization, while the director-level within-board exit hazard, which compares U.S.-connected directors with their boardmates, is the design least vulnerable to firm-level mean reversion. The lower panel states the cross-market prediction: the response is selective under intent-to-treat exposure and becomes indiscriminate once an explicit sanction renders the holding legally prohibited.

3.2 Data and Sources

We procure our director and resume data from two CSMAR modules: the Corporate Governance module (annual snapshots 2010–2024, 176K unique directors) and the director changes file (event-level, $\sim$1M rows). The annual file supplies resume text, education background, professional titles, and overseas work and education indicators, while the event file records appointment and departure dates alongside resignation-reason text; from these fields we construct the U.S.-nexus measure described above.

The remaining firm-level variables come from three further CSMAR modules. Following the standard governance-panel practice, we draw our financial controls—log assets, leverage (i.e., total liabilities over total assets), and ROA (i.e., net income over total assets)—from the Financial Statement modules, where they serve as time-varying covariates winsorized at the 1st and 99th percentiles. The Ownership and Control module supplies a time-varying state-owned/private (SOE/POE) classification together with top-10 shareholder concentration, while the board and compensation variables—board size (i.e., the board-of-directors count), the independent-director ratio, top-three executive pay, and aggregate institutional ownership—come from the Executive Compensation and Governance module, whose coverage exceeds 99% over the sample period.

Our estimation sample comprises 2,941 unique firms (1,109 treated and 1,832 control) observed over 2019–2024, yielding 17,171 firm-year observations. We classify a firm as treated if it had at least one U.S.-nexus director in the 2019–2020 baseline; control firms are matched by industry (China Securities Regulatory Commission, or CSRC, two-digit) and internationalization status, and the post-shock indicator equals one for years from 2021 onward.

3.3 Estimating Equations

To examine how the decoupling shock reshapes board composition, we estimate an exposure-based difference-in-differences (DiD) specification with firm and year fixed effects. For firm j in year t,

$$\text{U.S.-nexus share}_{jt} = \beta_1(\text{Treated}_j \times \text{Post}_t) + \gamma X_{jt} + \delta_j + \lambda_t + \epsilon_{jt}, \tag{1}$$

where Treated_j equals 1 for firms with at least one U.S.-nexus director in the 2019–2020 baseline, Post_t equals 1 for years $\geq$ 2021, and X_{jt} collects log assets, leverage, and ROA, while δ_j and λ_t represent firm and year fixed effects, respectively, and standard errors are clustered at the firm level. We expect the coefficient of interest, β_1, to be negative. A dose-response variant replaces Treated_j with the continuous pre-shock intensity $\overline{\text{U.S.-nexus share}}_{j,2019\text{–}2020}$, while an event-study variant replaces $\text{Treated}_j \times \text{Post}_t$ with a full set of year-by-treatment interactions.

The firm-level specification gives the magnitude of the board-composition change, yet, for reasons set out in the identification strategy below, it carries little causal weight, because its treatment is the pre-period level of the outcome. Our identifying design is therefore a director-level within-board exit hazard. For director i on the board of firm j in year t,

$$\text{Exit}_{ijt} = \beta_1(\text{U.S.-nexus}_i \times \text{Post}_t) + \beta_2\, \text{U.S.-nexus}_i + \theta_{jt} + \varepsilon_{ijt}, \tag{2}$$

where Exit_{ijt} indicates that the director leaves the board before year $t+1$, θ_{jt} denotes firm $\times$ year fixed effects, and standard errors are clustered by firm. The firm $\times$ year fixed effects restrict identification to the contrast between U.S.-connected directors and their non-U.S. colleagues on the same board in the same year, so that any firm-level trajectory is differenced out by construction; we expect $\beta_1 > 0$.

We estimate the replacement analysis on matched exit–entry pairs, regressing an indicator for domestic-technocrat replacement on Treated U.S. exit_{ijt} (i.e., a U.S.-nexus director leaving a treated firm post-shock), again with firm and year fixed effects. As Section 4 makes explicit, this analysis is descriptive: it characterizes who fills vacated seats and does not isolate a causal effect of director loss on replacement type. The Hong Kong analyses adopt the same DiD structure, with treatment defined by baseline Western-surname or U.S.-nationality directors, whereas the explicit-sanctions analysis uses a binary indicator for the eleven sanctioned firms.

3.4 Identification Strategy

The exposure difference-in-differences documents the size of the board-composition change. Because its treatment is the pre-period level of the outcome and the sanctions arrived as a single bundled episode, we assign causal weight to the within-firm and within-board-year comparisons—the placebo and the director-level exit hazard—which difference out any firm-level mean reversion by construction. The remainder of this subsection sets out the exposure logic we borrow as motivation, the mean-reversion threat the firm-level design faces, and the two within-unit designs that meet it.

Our identification strategy follows the exposure-based difference-in-differences framework, under which pre-period exposure defines treatment and an external shock supplies the source of quasi-exogenous variation. We define treatment by whether a firm had at least one U.S.-nexus director during the 2019–2020 baseline period, and we aim to identify the effect by exploiting the sanctions shock—driven by U.S. national-security considerations quasi-exogenous to individual firm board composition—as a shift in the costs facing U.S.-connected directors. This structure parallels the "China shock" literature, in which Autor et al. (2013) define treatment intensity by pre-period import exposure and locate identification in the quasi-exogeneity of the trade shock; Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022) provide formal treatments of identification in shift-share designs. Our setting meets the exogenous-shock condition, because the timing, scope, and targeting of U.S. sanctions were determined by Washington's national-security apparatus, independent of the board composition of individual Chinese firms. We borrow this exposure logic as motivation only. Although the license those shift-share designs provide to identify from shock exogeneity in place of parallel trends presumes many independent shocks, our setting rests on a single bundled sanctions episode whose instruments overlap in timing and targeting. With one shock, identification in a difference-in-differences turns on the comparability of treated and control trajectories, the parallel-trends condition that our treatment definition strains; shock exogeneity therefore supplies context for the treatment while leaving the parallel-trends burden in place. The shares themselves are plausibly endogenous to firm characteristics, and the firm and year fixed effects absorb time-invariant share-level confounders. We accordingly borrow the

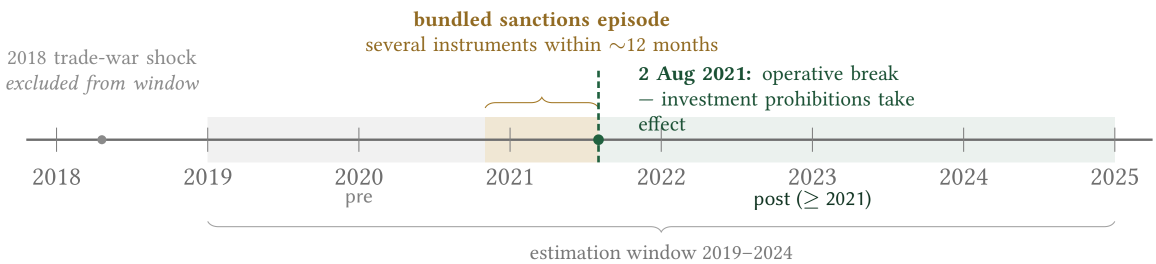

Figure 2: Timeline of the decoupling regime and the estimation window. The treatment is a bundle of U.S. measures that fired within roughly twelve months—Executive Order 13959 (November 2020), Executive Order 14032 (June 2021), the NS-CMIC list, the U.S. Persons Rule, and a contemporaneous tightening of the BIS Entity List—whose timing and targeting overlap, so the data identify their cumulative effect rather than any single instrument's. We date the treatment to the operative break of 2 August 2021, when the principal investment prohibitions took effect. The 2019–2024 estimation window excludes the distinct 2018 trade-war shock (the Section 301 tariffs and the first Entity List designations) and affords only a thin pre-period (2019–2020); because identification rests on the within-firm and within-board comparisons, that thin pre-period serves only as a descriptive check.

conceptual exposure-DiD logic of the China-shock literature while leaving its large-number-of-shocks asymptotic inference to settings with many independent shocks. This exposure-based design is aimed at the effect of the shock on the measured composition of the board—the outcome to which firms' U.S.-nexus exposure is mechanically linked—and we accordingly limit causal claims to that outcome, treating them as conditional on the identifying assumptions set out below.

The decoupling pressure we study operated as a regime composed of several legal instruments that fired in close succession: Executive Order 13959 (November 2020) and its broadening successor Executive Order 14032 (June 2021), the construction and expansion of the NS-CMIC list, the U.S. Persons Rule, and a contemporaneous tightening of the Bureau of Industry and Security (BIS) Entity List. Because these instruments overlap both in timing and in the firms they reach, we estimate the cumulative response to the bundle; the data cannot separate the marginal effect of Executive Order 14032 from that of the Persons Rule or the Entity List when all bind within a twelve-month window. We identify the cumulative reduced-form effect of the decoupling regime as a whole, dated to its operative break on August 2, 2021, when the principal investment prohibitions took effect. For the policy question that motivates this Article—how geopolitical fragmentation reshapes corporate boards—the bundle is the relevant treatment, because U.S.-connected directors faced the entire regime. We treat the earlier 2018 trade-war measures, namely the Section 301 tariffs and the first Entity List designations, as a distinct prior shock that the 2019–2024 estimation window deliberately excludes, so that the 2021 break is not conflated with trade-war effects.

Because we define treatment by the level of the outcome variable in the pre-period, treated and control firms are unlikely to share parallel pre-trends in that variable, since treated firms were accumulating U.S.-nexus directors throughout 2015–2019 while control firms had no comparable accumulation. The relevant identifying assumption therefore concerns the quasi-exogeneity of the sanctions shock to pre-period board composition: to establish the causal impact of the decoupling regime, the effect of this shock on board composition must arise solely through its influence on the costs facing U.S.-connected directors and be unrelated to any prior process by which Washington timed or targeted sanctions on the basis of which Chinese firms had Western-connected directors. The principal threat to this design is mean reversion. Because we select firms on a high pre-period U.S.-nexus level, a residual concern is that such firms might mechanically decline toward a common mean regardless of the shock; pre-trend parallelism is unattainable under this treatment definition and therefore cannot serve as the operative assumption. Under a counterfactual scenario where mean reversion drives the post-shock decline, one would expect the U.S.-nexus director share at high-baseline firms to fall back toward the average even absent any sanctions, and one would expect such a firm-level mechanical process to fall on every category of director at those firms. Our central defense against this rival reading is a within-firm placebo that compares U.S.-connected directors with the non-U.S. foreign directors at the same treated firms. Because firm-level mean reversion, mechanical turnover at the end of fixed board terms, and a market-wide de-internationalization trend would each depress all director types at a high-baseline firm, none of them can generate any asymmetry between U.S. and non-U.S. directors within the same firm-years; Section 4 reports this placebo (Table 10).

The resignation analysis embeds this within-firm comparison directly through a triple difference (U.S.-nexus $\times$ Post $\times$ Treated). Several further patterns examined in Section 4 bear on the same comparison: the behavior of the U.S.-nexus share in the single available pre-treatment year, the proportionality of post-shock exit to baseline exposure, and the composition of stated resignation reasons. The dose-response gradient warrants particular care, because a baseline-proportional decline is what mean reversion would also produce; such a gradient corroborates the shock channel without ruling reversion out, the task we reserve for the within-board hazard.[5]

The second, and our primary, defense is the director-level exit-hazard design introduced above, which holds firm and year jointly fixed. The unit of analysis is the director-year and the outcome is whether a director departs the board before the following year, while the specification absorbs firm $\times$ year fixed effects, so that the estimate compares each U.S.-connected director only with the non-U.S. colleagues serving on the same board in the same year. Any firm-level trajectory, including the mean-reverting drift of a high-baseline firm, is differenced out by construction, so that a firm-level pre-trend cannot produce the result. Section 4 reports the estimates from this design (Figure 3 and Table 2). A pre-shock pseudo-treatment built on a 2014–2015 baseline is infeasible in the present panel, which begins in 2018; the within-board design supersedes that check, because it removes by construction the firm-level selection the pseudo-treatment was meant to detect.

A separate question is whether recent critiques of two-way fixed-effects estimators bear on the design. The negative-weighting and contamination problems those critiques identify (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021; Borusyak et al., 2022) arise when treatment timing is staggered across units and treatment effects are heterogeneous, so that already-treated units act as controls for later-treated ones. Our setting is largely insulated from this concern, because the sanctions regime is common-timed: the operative break falls in 2021 for every treated firm, leaving no already-treated units to contaminate the compari-

[5] For the same reason—treatment defined by pre-period outcome levels—synthetic control methods (Abadie, 2021) and synthetic difference-in-differences (Arkhangelsky et al., 2021) are inapplicable, as the reweighting procedure cannot match treated firms' pre-treatment trajectories from control units with near-zero baseline exposure.

son, so that the design reduces to a common-timing two-group difference-in-differences outside the staggered case those critiques target. The within-board hazard inherits the same protection, since its firm $\times$ year fixed effects compare directors within a single firm-year and require no cross-cohort comparison. For the continuous dose-response specification, in which treatment intensity varies across firms, Callaway et al. (2024) show that level effects are identified under a generalized parallel-trends condition while the dose gradient itself requires stronger assumptions, so we read the gradient as corroborative evidence that carries no independent identifying force. Should the firm-level Entity List designation dates prove to generate genuinely staggered timing—a refinement we take up in the Discussion—a heterogeneity-robust estimator in the manner of Callaway and Sant'Anna (2021) would be the appropriate response.

We restrict the estimation sample to 2019–2024 to isolate the primary sanctions escalation beginning in 2021 from earlier trade-war measures. The U.S.–China trade conflict escalated in stages: initial Entity List designations affected approximately 140 firms from 2018, followed by a relative pause in 2019–2020, and then the major sanctions escalation via Executive Order 14032 in June 2021 and subsequent expansions. Including 2015–2018 in the pre-period would conflate two distinct policy shocks—the early trade-war skirmishes and the later sanctions regime—and so would complicate identification of the 2021 break. The 2019–2024 window provides a brief pre-period during which the early trade-war effects had settled and the major sanctions had not yet taken effect. Shock-timing clarity supplies the rationale for this restriction; identification is carried by the within-unit designs above, with the thin pre-period serving only as a descriptive check.

The Hong Kong cross-market extension affords an additional pre-trend check. Because board internationalization began later among Hong Kong-listed firms than among mainland A-share companies, the Hong Kong panel affords a longer pre-treatment window than the mainland data. The mainland and Hong Kong event studies appear in Appendix A.3 (Figures A1 and A2).

Our causal interpretation is bounded by where the design's identifying power lies, which is strongest for board composition, the outcome to which firms' pre-period U.S.-nexus exposure is mechanically and contemporaneously linked. We therefore treat the director-exit results, together with the placebo decomposition that isolates the U.S.-specific component from a concurrent market-wide de-internationalization, as our most defensible causal interpretation, while acknowledging the residual threats to identification set out below. Downstream firm-level outcomes such as valuation and related-party activity lie further from the source of variation, since any effect of the shock on them would operate through multiple channels—direct sanctions exposure, customer and supplier disruption, financing conditions, and investor sentiment—that the exposure measure does not separate. We accordingly report the replacement and compensation patterns as descriptive associations conditional on the identified exit, leaving the causal effect of director loss on those outcomes unestimated. Figure 1 summarizes this identification logic and the underlying mechanism.

3.5 Summary Statistics

Table 1: Summary Statistics

Panel A: Firm-Year Variables (2019–2024)						
	N	Mean	SD	P25	Median	P75
U.S.-nexus director share	17,171	0.021	0.038	0.000	0.000	0.044
U.S.-nexus directors (count)	17,171	0.403	0.713	0.000	0.000	1.000
Foreign-director share	17,171	0.018	0.055	0.000	0.000	0.000
Board size	17,110	8.41	1.78	7.0	9.0	9.0
Independent-director ratio (%)	17,109	38.13	5.72	33.33	36.36	42.86
Log assets	17,171	22.73	1.631	21.62	22.47	23.56
Leverage	17,171	0.462	0.272	0.287	0.444	0.606
ROA	17,171	0.019	0.157	0.006	0.030	0.063
Top-3 executive pay (M¥)	17,088	4.32	4.56	2.09	3.08	4.93
Institutional ownership (%)	17,088	44.49	24.78	24.83	45.27	64.21
State-owned enterprise	17,171	0.367	—	—	—	—

Panel B: Treatment and Control Breakdown						
	Treated (1,109 firms)			Control (1,832 firms)		
	N	Mean	SD	N	Mean	SD
U.S.-nexus director share	6,433	0.053	0.044	10,738	0.002	0.010
Board size	6,429	8.61	1.89	10,681	8.29	1.70
Independent-director ratio (%)	6,432	38.10	5.78	10,677	38.14	5.69
Top-3 executive pay (M¥)	6,419	4.78	5.35	10,669	4.04	3.98
Institutional ownership (%)	6,427	44.89	24.77	10,661	44.25	24.79

Panel C: Sample Composition
2,941 unique firms (1,109 treated, 1,832 control); 17,171 firm-years
Pre (2019–2020): 5,679 firm-years; Post (2021–2024): 11,492 firm-years
SOE firms: ∼1,026 (6,306 firm-years); POE firms: ∼2,149 (13,502 firm-years)

Note: This table reports firm-year summary statistics for the preferred mainland A-share sample over 2019–2024 and the associated treatment-control breakdown.

Table 1 presents our summary statistics for the preferred sample of 17,171 firm-year observations spanning 2019–2024. Panel A reports firm-year-level variables. The average board has 8.4 directors (board of directors only, excluding the supervisory board and executives), with U.S.-nexus directors comprising 2.1% of the board on average. The mean independent-director ratio is 38.1%, just above the regulatory minimum of one-third, with 44% of firm-years bunched at exactly 33.3%. Top-3 executive compensation averages 4.32 million yuan, and aggregate institutional ownership is 44.5%, while state-owned enterprises comprise 36.7% of firm-years. Panel B breaks down key variables by treatment status: treated firms have higher baseline U.S.-nexus shares (5.3% vs. 0.2%), slightly larger boards, and higher executive compensation.

4 The Exit of U.S.-Nexus Directors

We read the broad exit of U.S.-connected directors documented in this section as consistent with compliance chill, the channel through which the U.S. Persons Rule and the surrounding sanctions framework raised the personal legal and reputational cost of board service at politically exposed Chinese firms. The exit appears across firms facing no direct legal prohibition, concentrates among independent directors, and operates through passive non-renewal at term end, and each of these features maps onto the legal structure set out in Section 2. The concentration among independent directors follows from their fee- and equity-based compensation falling within the regulations' reach and from their lower switching costs; the appearance of the exit even at firms under no firm-specific prohibition follows from liability attaching personally to the U.S. person; and the clustering of departures at scheduled term end, with mid-term resignation playing little role, is the low-friction route by which a board sheds exposure once the legal cost of continued service rises. Note that the design identifies the director-exit outcome while leaving the legal-liability channel indirectly observed. We therefore present compliance chill as the reading most consistent with the assembled evidence, while acknowledging that the data do not adjudicate it against alternatives such as reputational avoidance unrelated to legal liability or anticipatory firm-side board management.

4.1 The Within-Board Exit Hazard

Our primary identifying design, specified in Section 3, is the director-level exit hazard with firm $\times$ year fixed effects. The unit of analysis is the director-year, the outcome is whether a director departs the board before the following year, and the firm $\times$ year fixed effects restrict identification to the contrast between each U.S.-connected director and the non-U.S. colleagues serving on the same board in the same year. Because any firm-level trajectory, including the mean-reverting drift of a high-baseline firm, is differenced out by construction, a firm-level pre-trend cannot generate the estimate.

Figure 3 plots the within-board exit gap by year, and the path it traces shows no positive pre-trend. The 2019 coefficient is small and not significant at the 10% level ($+1.6$pp), while the 2018 coefficient is significantly negative (-3.1pp), the opposite sign from the post-shock gap. Once the sanctions bind, the gap rises monotonically, reaching $+2.0$pp in 2021, $+3.6$pp in 2022 (significant at the 1% level), and $+4.5$pp in 2023 (significant at the 1% level), against a mean annual exit rate near 14.5%.

Table 2 reports the underlying estimates. Column 1 is a pooled linear-probability model; column 2 adds firm and year fixed effects; columns 3–5 absorb firm $\times$ year fixed effects, which restrict identification to the within-board contrast and difference out any firm-level mean reversion. The coefficient on U.S.-nexus $\times$ Post is positive in every specification, reaching $+3.0$pp (significant

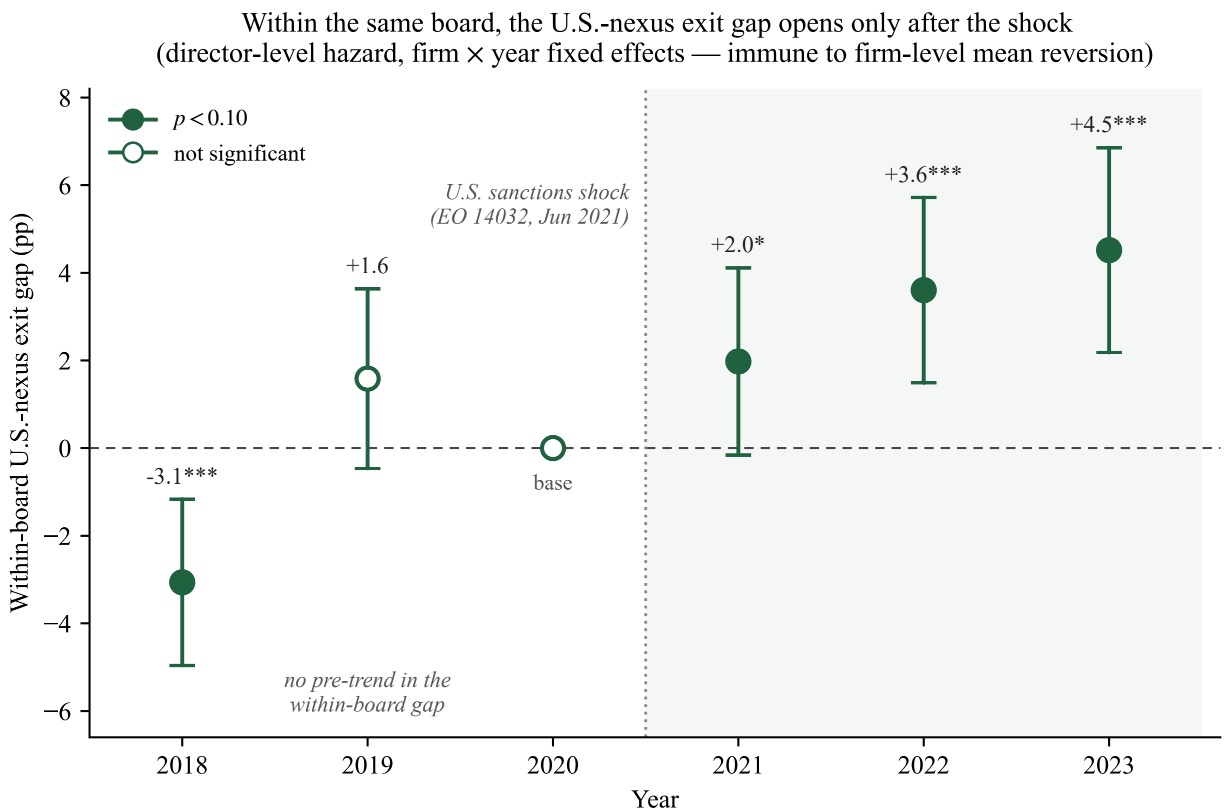

Figure 3: Within-board exit hazard for U.S.-nexus directors, by year. Each point is the coefficient on U.S.-nexus × year in a director-level linear-probability model of board exit with firm × year fixed effects (base year 2020), estimated on 323,098 director-years across 2,941 firms; whiskers are firm-clustered 95% confidence intervals (filled markers denote $p < 0.10$). Because the firm × year fixed effects absorb every firm-level confounder, the coefficient compares U.S.-connected directors only with their non-U.S. colleagues on the same board in the same year, so a firm-level pre-trend or mean-reverting drift cannot generate it. The within-board gap shows no positive pre-trend before 2021 and climbs monotonically after the sanctions bind; the body text and Table 2 report the year-by-year and pooled estimates.

at the 1% level) over the full 2018–2023 window in column 4 and $+3.8$pp (significant at the 1% level) within treated firms in column 5; it attenuates to $+1.5$pp (not significant at the 10% level) only in the narrow 2019–2023 window, where the growing dynamic effect is averaged against a near-zero 2020 base. The positive U.S.-nexus main effect measures the pre-period within-board exit gap, which Figure 3 shows to carry no positive pre-trend before 2021.

Table 2: Director-Level Exit Hazard with Firm × Year Fixed Effects

	(1) Pooled	(2) Firm + Year FE	(3) Firm × Year FE	(4) Firm × Year (2018–23)	(5) Firm × Year (treated)
U.S.-nexus × Post	+0.0122	+0.0118	+0.0146	+0.0295***	+0.0379***
	(0.228)	(0.243)	(0.146)	(0.001)	(<0.001)
U.S.-nexus	+0.0178**	+0.0213***	+0.0186***	+0.0037	+0.0186***
	(0.014)	(0.003)	(0.010)	(0.513)	(0.010)
Firm FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Firm × Year FE	No	No	Yes	Yes	Yes
Window	2019–23	2019–23	2019–23	2018–23	2019–23
Sample	All	All	All	All	Treated
Mean exit rate	0.145	0.145	0.145	0.144	0.149
Director-years	272,817	272,817	272,817	323,098	105,309
Firms (clusters)	2,934	2,934	2,934	2,941	1,107

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The dependent variable is an indicator that a director leaves the board between year t and $t+1$; director-years whose firm is not listed in $t+1$ are dropped so that delistings are not counted as exits. Post $= 1$ for $t \geq 2021$. Columns 3–5 absorb firm × year fixed effects, so identification rests on the within-board contrast between U.S.-nexus and non-U.S. directors; column 4 uses the full 2018–2023 window and column 5 restricts the sample to treated firms. Standard errors are clustered by firm.

Scheduled term cycles are an unlikely explanation for the gap, since routine turnover falls on a director's U.S. and non-U.S. board colleagues alike and is absorbed by the fixed effects. We cannot yet rule out, however, that U.S.-connected directors sat systematically closer to term expiry than their boardmates, a possibility a director-tenure control would address (Section 7).

4.2 The Firm-Level Exit

Table 3: Difference-in-Differences: US-Nexus Director Share

	(1) Preferred 2019–2024 Post≥2021	(2) Original 2018–2024 Post≥2022	(3) Early Post 2019–2024 Post≥2020	(4) Extended 2018–2024 Post≥2021
DV: U.S.-nexus director share				
Treated × Post	-0.0200^{***}	-0.0200^{***}	-0.0156^{***}	-0.0176^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Alternative DVs (Column 1 specification only)				
DV: U.S.-nexus directors (count)	$\hat{\beta} = -0.3692^{***}$, $p < 0.001$			
DV: Any U.S.-nexus director (0/1)	$\hat{\beta} = -0.3386^{***}$, $p < 0.001$			
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	17,171	19,806	17,171	19,806
R^2 (within)	0.0504	0.0494	0.0183	0.0374

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The main panel estimates difference-in-differences effects on the U.S.-nexus director share across sample windows, with controls, firm fixed effects, year fixed effects, and firm-clustered inference.

Table 3 reports our base difference-in-differences estimates of the geopolitical shock on U.S.-nexus board representation. In our preferred specification (Column 1, the 2019–2024 sample with Post $\geq$ 2021), the U.S.-nexus $\times$ Post coefficient is $\hat{\beta} = -0.020$ and is statistically significant at the 1% level, which indicates that treated firms experienced a 2.0 percentage-point decline in U.S.-nexus director share relative to control firms. Read against a treated-firm pre-shock mean of approximately 5.9%, this amounts to a one-third reduction. The estimate is notably stable across sample windows. Extending the panel to 2018–2024 in Column 2, which also shifts the post cutoff to 2022, yields an identical point estimate; holding the 2021 cutoff fixed over that window leaves it nearly unchanged (Column 4, -0.018^{***}); and moving the post cutoff earlier to 2020 in Column 3 produces a somewhat smaller but still highly significant effect (-0.016^{***}). Under industry $\times$ year fixed effects the estimate is virtually unchanged (-0.020^{***}), which confirms that the result does not reflect sector-wide trends. Even under two-way firm and year clustering, the most demanding standard-error specification we report, the coefficient remains significant at the 1% level despite a fourfold increase in the standard error.

Table 4: Event Study: Year-by-Treatment Interactions

	(1) Full Sample 2018–2024 Base = 2020	(2) Preferred 2019–2024 Base = 2019
DV: U.S.-nexus director share		
Treated × 2018	-0.0075^{***}	
	($<$0.001)	
Treated × 2019	-0.0004	
	(0.719)	
Treated × 2020		$+0.0003$
		(0.763)
Treated × 2021	-0.0086^{***}	-0.0083^{***}
	($<$0.001)	($<$0.001)
Treated × 2022	-0.0160^{***}	-0.0158^{***}
	($<$0.001)	($<$0.001)
Treated × 2023	-0.0244^{***}	-0.0242^{***}
	($<$0.001)	($<$0.001)
Treated × 2024	-0.0321^{***}	-0.0319^{***}
	($<$0.001)	($<$0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm
Observations	19,806	17,171

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports year-by-treatment interactions for the U.S.-nexus director share, with base years and sample windows shown in the column headers.

Table 4 presents our event-study coefficients. In the preferred 2019–2024 specification (Column 2, with base year 2019), the single available pre-treatment coefficient, that for 2020, is $+0.0003$ and not significant at the 10% level, showing no detectable pre-shock divergence in the one pre-treatment year the window affords. The post-shock coefficients decline monotonically thereafter: -0.008^{***} (2021), -0.016^{***} (2022), -0.024^{***} (2023), and -0.032^{***} (2024). The full 2018–2024 specification (Column 1, with base year 2020) reveals a significant 2018 coefficient (-0.0075^{***}) alongside a clean 2019 reading (-0.0004, not significant at the 10% level). We regard the 2018 anomaly as an isolated blip attributable to the U.S.–China trade war, since it is smaller than every post-treatment coefficient and, as we show in the Discussion, is driven by a small set of early-treated firms and disappears once they are excluded. We accordingly adopt the 2019–2024 window as our primary specification.

Table 5: Dose-Response and Position-Type Concentration

Panel A: Extension 1a — Dose-Response by Position Type		
	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)
Intensity × Post	-0.6966^{***}	-0.1173^{***}
	($<$0.001)	($<$0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm
Observations	17,171	17,171

Panel B: Extension 1c — Position Type Split (Binary Treatment)			
	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)	(3) Independent-director share
Treated × Post	-0.0477^{***}	-0.0063^{***}	+0.0017
	($<$0.001)	($<$0.001)	(0.670)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustering	Firm	Firm	Firm
Observations	17,171	17,171	10,809
R^2 (within)	0.0503	0.0107	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels compare dose-response and binary-treatment estimates by director position type; the Panel B independent-to-non-independent coefficient ratio is 7.5×.

Table 5 reports two extensions. Panel A presents the dose-response gradient by position type, which is 6× steeper for independent directors (-0.697^{***}) than for non-independent directors (-0.117^{***}). Panel B reports the binary difference-in-differences by position type: -0.048^{***} for independent directors against -0.006^{***} for non-independent directors, a 7.5× ratio. We read this concentration among independents as consistent with the U.S. Persons Rule bearing on advisory relationships and with independent directors facing lower switching costs. Notably, the board independence ratio itself is unaffected ($+0.002$, not significant at the 10% level), which confirms that firms replaced departing U.S.-nexus independents with domestic independents and maintained board structure.

Table 6: Resignation Reasons as Quasi-exogeneity Evidence

Panel A: Euphemistic Resignation Rates (2×2)				
	Pre (2019–20)	Post (2021–24)	Diff	
U.S.-nexus	35.5% (n=403)	31.3% (n=838)	−4.2 pp	
Non-U.S.-nexus	38.0% (n=25,392)	38.6% (n=53,686)	+0.6 pp	
DiD $= -4.8$ pp, $\chi^2 = 22.03$, $p = 0.0001$				

Panel B: Linear probability model (DV: euphemistic resignation)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Triple-Diff
U.S.-nexus × Post	-0.0666^{**}	-0.0652^{**}	-0.0701^{**}	
	(0.027)	(0.030)	(0.022)	
U.S.-nexus × Post × Treated				-0.1221^{**}
				(0.011)
U.S.-nexus × Post				$+0.0434$
				(0.347)
U.S.-nexus × Treated				-0.0974^{**}
				(0.035)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Observations	54,117	54,117	54,117	54,117

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports resignation-reason evidence using euphemistic resignation rates and linear probability models for U.S.-nexus departures before and after 2021.

Table 6 examines the stated reasons for resignation. Against the naive prediction that euphemistic reasons would increase, the post-shock U.S.-nexus departures show a decrease: $\hat{\beta}$(U.S.-nexus $\times$ Post) $= -0.065$ (significant at the 5% level). The triple interaction with Treated is stronger still (-0.122, significant at the 5% level). Decomposing the reasons reveals the mechanism, since "personal reasons" declined (-0.077, significant at the 1% level) while "term expiry" increased ($+0.061$, significant at the 10% level), and the effect concentrates among independent directors (-0.123, significant at the 1% level). Taken together, these patterns indicate that U.S.-nexus exits occurred through passive non-renewal at term end, with firms waiting for natural board renewal cycles and forced mid-term resignations playing little role, which is precisely the departure pattern a chilling-cost mechanism predicts.

4.3 Board Reconstitution: Who Replaces the Departing Directors

In this section we characterize who fills the seats vacated by U.S.-nexus directors. Having interpreted the exit as the likely causal effect of the shock, subject to the threats to identification discussed in Section 7, we now turn to how boards recompose themselves around it. We treat this second step as associational, because it conditions on the identified exit and describes the composition of replacements without asserting a separate causal effect, since replacement type is jointly determined with the firm's broader response to the shock.

Figure 4 sets the scene. At treated firms the U.S.-nexus director share falls by roughly a third after 2021, while non-U.S. foreign representation and overall board size hold steady, showing that the vacated seats are refilled; the remainder of this section asks who fills them.

Table 7: Replacement-Director Professionalization

Panel A: DV = Domestic technocrat (Replacement-Pair Level)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Full
Treated (U.S.-nexus exit)	$+0.0282^{**}$	$+0.0411^{***}$	$+0.0108^{**}$	$+0.0108^{**}$
	(0.014)	($<$0.001)	(0.025)	(0.024)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	No	No	Yes
Observations	314,050	314,050	314,050	314,050
R^2	0.0001	0.0025		

Panel B: DV = CCP-affiliated replacement (Preferred Specification)	
	(1)
Treated (U.S.-nexus exit)	-0.0012
	(0.765)
Year FE	Yes
Firm FE	Yes
Controls	Yes
Observations	314,050

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses replacement-pair data to estimate domestic technocrat and CCP replacement outcomes for seats vacated by U.S.-nexus exits.

Table 7 examines the directors who step into the seats that departing U.S.-nexus directors leave behind. Drawing on 314,050 matched exit–entry replacement pairs, our preferred specification (Column 4) yields $\hat{\beta}(\text{Treated U.S. exit}) = +0.011$ (significant at the 5% level), so that U.S.-nexus exits from treated firms post-shock are associated with a 1.1 percentage-point higher rate of

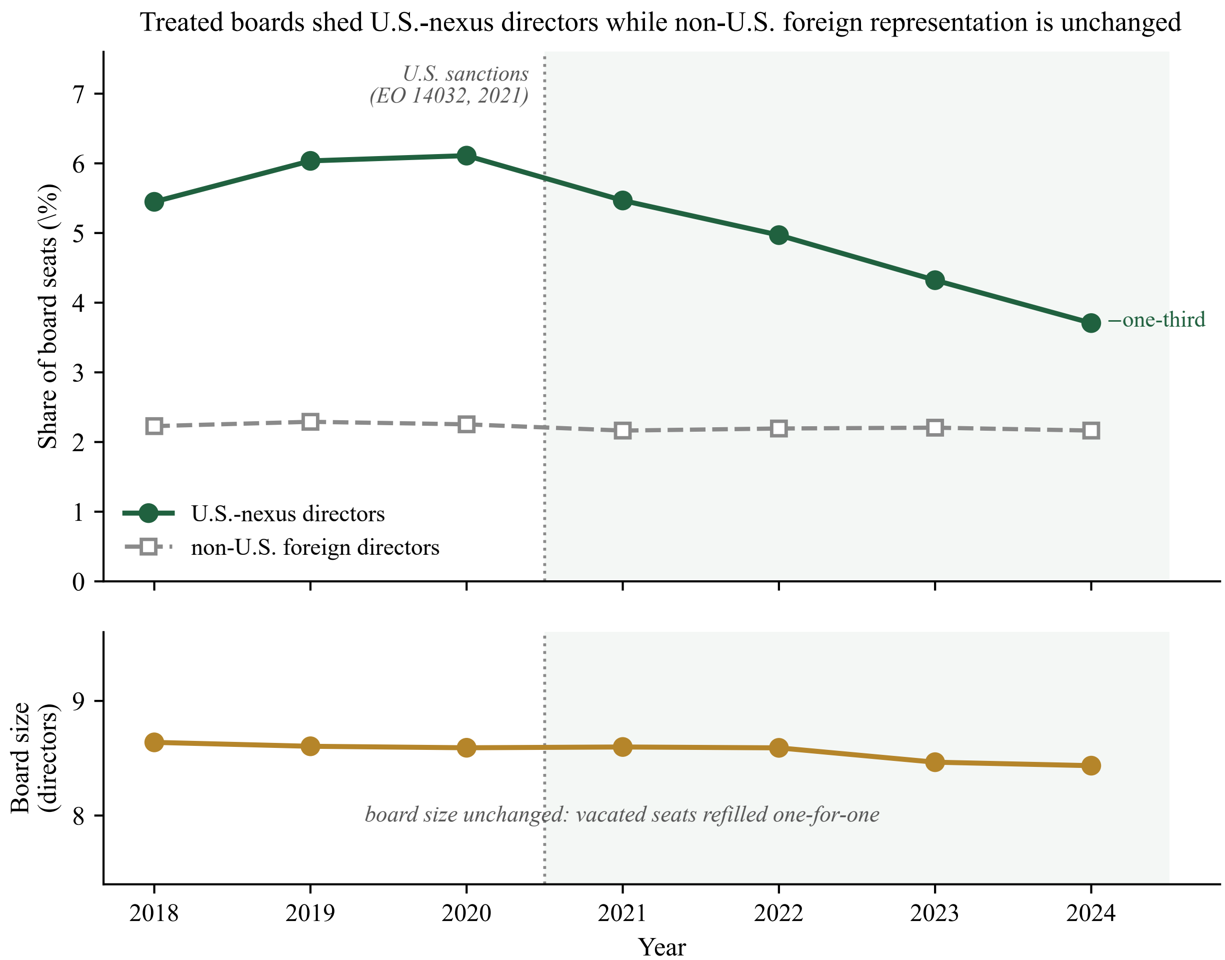

Figure 4: Board composition at treated mainland firms, 2018–2024. Treated firms are those with at least one U.S.-nexus director in the 2019–2020 baseline; means are unweighted across treated firms. Panel A plots the mean share of board seats held by U.S.-nexus directors and by non-U.S. foreign directors; Panel B plots mean board size (the CSMAR board-of-directors count). After the 2021 sanctions the U.S.-nexus share falls by about a third (from roughly 6% to 3.7%), whereas non-U.S. foreign representation is flat—the U.S.-specific pattern that the within-firm placebo formalizes (Table 10). Board size is unchanged throughout, so the vacated seats are refilled one-for-one; the replacement analysis (Table 7) shows the replacements are disproportionately domestic technocrats.

replacement by domestic technocrats. We read this association as a descriptive pattern in who fills the vacated seats. The CCP-replacement rate, by contrast, exhibits no differential change (-0.001, not significant at the 10% level), a clean descriptive non-result that runs directly against a state-capture narrative.

Table 8 reports three extensions. Panel A shows that the composite technical-background index is higher for replacements of U.S.-nexus exits ($+0.006$, significant at the 5% level), a movement driven entirely by the professional-credential component ($+0.014$, significant at the 1% level). Panel B turns to board size: CSMAR's pre-computed board size (board of directors only, mean 8.4) registers no change ($+0.009$, not significant at the 10% level), consistent with complete one-for-one replacement. The previously reported shrinkage (-0.32, significant at the 5% level) was, notably, an artifact of the director-panel count, which erroneously included supervisory-board members and executives (mean 19.1). The CSMAR independent-director ratio is likewise unchanged (-0.086pp, not significant at the 10% level), which confirms that the independence ratio was maintained. Panel C examines compensation: independent-director pay at privately owned enterprises rose by 16% ($+0.160$, significant at the 5% level). We read this as an associational pattern consistent with upward pressure on independent-director pay following the exit of U.S.-connected directors, although it may also reflect a general post-shock repricing of director risk, and we do not interpret it as an identified causal effect. Top executive compensation is null overall (-0.014, not significant at the 10% level), while SOE executive pay declined by 5.7% (significant at the 5% level), a different labor market that is likely shaped by administrative pay constraints. The Appendix consolidates robustness across all specifications, including alternative fixed-effect structures, clustering levels, and wild cluster bootstrap inference (Table A7).

The directors who depart are more highly credentialed than those who remain. Table 9 compares U.S.-connected directors with their non-U.S. colleagues on two credentials that lie outside the U.S.-nexus definition, measured in the 2019–2020 baseline. U.S.-connected directors hold a graduate degree at nearly twice the domestic rate (83% against 45%), a gap of 34 percentage points that survives when each U.S.-connected director is compared only with the non-U.S. colleagues on the same board in the same year ($+0.340$, significant at the 1% level). On senior domestic professional credentials the gap is smaller, and among independent directors—the group through which the exit runs—it narrows to a within-board difference that is not significant at the 10% level ($+0.033$). The departing directors thus carried distinctly more graduate-level and internationally oriented human capital, while the domestic professionals who remained and stepped into the vacated seats were their equals on domestic professional standing. This reading anticipates the consequence evidence of Section 6, where the loss of these directors leaves no detectable near-term governance cost.

Table 8: Replacement: Credentials, Board Size, and Compensation

Panel A: Extension 2a — Technical-background index and Components

	(1)	(2)	(3)	(4)	(5)
DV:	Technical-background index	Graduate degree	Professional credential	Domestic only	Non-CCP
Treated (U.S.-nexus exit)	$+0.0058^{**}$	$+0.0052$	$+0.0143^{***}$	$+0.0024$	$+0.0012$
	(0.011)	(0.359)	(0.002)	(0.476)	(0.765)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	314,050	314,050	314,050	314,050	314,050

Panel B: Extension 2b — Board Size and Independence Ratio

	(1)	(2)	(3)
DV:	Board size	Independent-director ratio	Director count (incl. supervisors)
Source:	CSMAR comp/gov	CSMAR comp/gov	Director panel
Treated $\times$ Post	$+0.0090$	-0.0859	-0.3183^{**}
	(0.821)	(0.611)	(0.013)
Firm FE + Year FE + Controls	Yes	Yes	Yes
Observations	17,110	17,109	17,171

Panel C: Extension 2c — Compensation

	(1)	(2)	(3)	(4)	(5)
DV:	Log(indep. director pay)	Log(indep. director pay)	Log(indep. director pay)	Log(top-3 exec. pay)	Log(top-3 exec. pay)
Sample:	Full	POE only	SOE only	Full	SOE only
Treated $\times$ Post	$+0.1009$	$+0.1596^{**}$	-0.0364	-0.0142	-0.0574^{**}
	(0.107)	(0.028)	(0.778)	(0.357)	(0.022)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	7,425	5,437	1,988	17,075	5,467

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Board size counts only the board of directors (mean 8.4); the director count also includes supervisors and executives (mean 19.1), so the apparent shrinkage in that count is an artifact. Independent-director pay is measured at the director level; top-three executive pay is measured at the firm level.

Table 9: Director Credentials: U.S.-Connected versus Domestic Directors

Panel A: All directors				
	(1) Domestic	(2) U.S.-nexus	(3) Gap (pp)	(4) Within-board
Graduate degree (Master's/PhD/MBA)	45.4	83.1	+37.7	$+0.340^{***}$
Senior professional title	33.6	46.3	+12.6	$+0.155^{***}$
Panel B: Independent directors only				
	(1) Domestic	(2) U.S.-nexus	(3) Gap (pp)	(4) Within-board
Graduate degree (Master's/PhD/MBA)	66.0	88.1	+22.0	$+0.212^{***}$
Senior professional title	54.1	56.9	+2.9	+0.033

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The sample is the mainland director panel in the 2019–2020 pre-shock baseline. Columns 1 and 2 report the share of directors holding each credential, in percent, for non-U.S. (domestic) and U.S.-nexus directors respectively; column 3 is the percentage-point gap; column 4 is the coefficient on U.S.-nexus in a linear-probability model of the credential indicator with firm $\times$ year fixed effects and standard errors clustered by firm, so that it compares each U.S.-connected director only with the non-U.S. colleagues on the same board in the same year. Graduate degree is a CSMAR education code of master's, doctorate, or MBA/EMBA; senior professional title flags senior-engineering, professorial, or accountancy (CPA) and equivalent designations in the CSMAR professional-title field. Neither credential enters the U.S.-nexus treatment definition. The within-board estimates are identified from the 1,909 baseline firm-years that contain both U.S.-connected and non-U.S. directors.

4.4 Isolating the U.S.-Specific Channel

A central question for our identification strategy is whether the director exit we document reflects U.S.-specific sanctions pressure or a broader de-internationalization of corporate boards. To address it, we report placebo designs across both markets in Table 10, and the resulting pattern furnishes the paper's most credible secondary check on identification. Under a counterfactual scenario in which market-wide de-internationalization drives the exit, one would expect non-U.S. foreign directors to leave treated firms alongside their U.S.-connected colleagues; the placebos test that prediction directly.

The within-firm placebo in columns 1–3 holds the firm fixed. At mainland treated firms, non-U.S. foreign directors are entirely unaffected ($+0.001$, not significant at the 10% level), so that the exit is confined to the U.S.-nexus group. At Hong Kong intent-to-treat firms, by contrast, non-U.S. foreign directors increased ($+0.010$, significant at the 5% level), a movement consistent with substitution toward non-U.S. internationals. At explicitly sanctioned firms, non-U.S. foreign directors declined (-0.029, significant at the 10% level), which suggests that formal legal sanctions create a broader reputational deterrent reaching all foreign directors.

Column 4 reports an unexposed-firm placebo. Firms that held non-U.S. foreign directors but no U.S. directors in the baseline experienced a marked decline in their own-type directors (-0.024,

Table 10: Non-US Foreign-Director Response: Cross-Market Placebo Tests

	(1) Mainland	(2) HK (ITT)	(3) HK (Sanctioned)	(4) HK (Placebo T2)
Panel A: US/Western Director Exit (Actual Effect)				
Treated × Post	-0.0200^{***}	-0.0131^{**}	-0.0135	-0.0001
	(<0.001)	(0.012)	(0.309)	(0.894)
Treatment definition	U.S.-nexus	U.S.-nationality	Sanctioned	Other foreign
DV	U.S.-nexus share	U.S.-nationality share	U.S.-nationality share	U.S.-nationality share
Panel B: Non-US Foreign Directors at Treated Firms (Placebo)				
Treated × Post	$+0.0007$	$+0.0101^{**}$	-0.0294^{*}	-0.0236^{***}
	(0.484)	(0.034)	(0.053)	(<0.001)
DV	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share
Treated firms	1,109	137	11	519
Control firms	1,841	2,515	2,652	1,595
Observations	17,171	21,566	21,612	19,301
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Columns 1–3 define treatment by baseline U.S.-nexus, U.S.-nationality, or sanctions exposure; column 4 defines placebo treatment as at least one non-U.S. foreign director in the baseline and zero U.S. directors.

significant at the 1% level), revealing a concurrent market-wide de-internationalization in Hong Kong boards that operates independently of U.S. sanctions. The event study displays clean pre-trends in 2018–2019 and a monotonic decline after 2021, a profile that points to the National Security Law, COVID-era border restrictions, and the wider Sino-Western tensions as contributors to foreign-director attrition across the market. An analogous test on the mainland, among firms with non-U.S. foreign directors but no U.S.-nexus exposure, likewise yields a significant decline (-0.016, significant at the 1% level; Table A6).

Taken together, the cross-market pattern supports three conclusions. First, the evidence points to a genuine U.S.-specific channel, because at treated firms U.S. directors exit while non-U.S. foreign directors are unaffected on the mainland and increase in Hong Kong, a pattern that diverges from the de-internationalization counterfactual. Second, a distinct de-internationalization channel operates on different firms and through a different mechanism. Third, the surname-based estimate (-0.025***) captures both channels, whereas the nationality-based estimate (-0.013**) isolates the U.S.-specific component, and the gap between the two is consistent with the market-wide trend we observe at unexposed firms. Appendix A.3 reports robustness to alternative treatment definitions.

5 Cross-Market Replication: Hong Kong

The mainland results establish that exposed firms shed U.S.-nexus directors after the shock, with the within-board hazard supplying the causal anchor, and they document the domestic professionalization of the replacement directors that followed. A natural question is whether these effects generalize beyond the A-share universe. The Hong Kong Stock Exchange hosts a distinct population of Chinese firms, many of them internationally oriented, which are subject to different regulatory and disclosure regimes and which report quarterly ownership breakdowns that are unavailable in A-share data. In this section we extend the analysis to this independent sample to test whether the same shock produced parallel director exits and how board substitution differs across the two markets; the capital-base response, which surfaces only among the explicitly sanctioned firms, is reported in Appendix A.2.

5.1 Foreign-Director Exit in Hong Kong

Table 11: Foreign-Director Exit: Hong Kong-Listed Chinese Firms

	(1)	(2)	(3)	(4)
	DV: Western-foreign director share		DV: Foreign-dialect director share	
Panel A: Baseline DiD				
Treated × Post	-0.0274^{***}	-0.0253^{***}	-0.0260^{***}	-0.0224^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Panel B: Dose-Response				
Intensity × Post	-0.2695^{***}	-0.2449^{***}	-0.1199^{***}	-0.1092^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Panel A statistics				
Observations	23,869	21,612	23,869	21,612
Firms	2,606	2,535	2,606	2,535
R^2 (within)	0.0450	0.0418	0.0214	0.0183
Panel B statistics				
R^2 (within)	0.0972	0.0884	0.0380	0.0349

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels report baseline difference-in-differences and dose-response estimates for Hong Kong-listed Chinese firms using Western-surname and foreign-dialect director-share outcomes.

Table 11 replicates the mainland exit result in an independent panel of Hong Kong-listed Chinese firms (roughly 2,600 firms over 2014–2025, with estimation samples that are somewhat smaller after controls and coverage restrictions, as reported in the table). We define treatment as having ≥ 1 Western-surname director in the 2019–2020 baseline, with Post $\geq$ 2021. Panel A reports the baseline DiD. The Western-surname measure yields $\hat{\beta} = -0.025$, an estimate that is significant at the 1% level, so that treated firms experienced a 2.5 pp decline in Western director share, a magnitude that closely parallels the mainland 2.0 pp figure. The broader foreign-dialect measure produces a similar effect (-0.022, significant at the 1% level), and Panel B confirms a dose-response gradient (-0.245, significant at the 1% level).

An independent nationality-based classifier decomposes this effect. Defining treatment as ≥ 1 U.S.-nationality director in the 2019–2020 baseline (137 firms, excluding the 11 sanctioned) and using the share of U.S.-nationality directors as the dependent variable, we obtain an intent-to-treat estimate of -0.013, which is significant at the 5% level (Table 10). The gap between the surname estimate (-0.025) and the nationality estimate (-0.013) quantifies the non-U.S. Western component of the exit, since approximately half the surname-based effect reflects the departure of directors who are Western-aligned but not U.S. nationals, a pattern consistent with a broader de-internationalization trend distinct from the U.S. sanctions. Because both measures rest on nationality-defined treatment, the nationality estimate provides the most direct comparison with the mainland figure.[6]

At intent-to-treat firms, the share of non-U.S. foreign directors rose post-shock ($+0.010$, significant at the 5% level), a movement consistent with boards offsetting U.S.-connected departures by recruiting other international directors, although the design cannot confirm that the recruitment was a deliberate substitution response. This pattern stands apart from the mainland one, in which firms replaced U.S.-nexus directors with domestic technocrats. The divergence plausibly reflects differing labor-market depths, because the Hong Kong exchange hosts a deep pool of non-U.S. international professionals (the baseline non-U.S. foreign share is 5.6%), whereas mainland A-share boards draw from a domestic talent market.

At the eleven explicitly sanctioned firms, the pattern differs. Non-U.S. foreign directors also declined (-0.029, significant at the 10% level), even though these directors carried no direct U.S. compliance exposure. Drawn from a small set of firms, this marginal estimate is consistent with explicit sanctions generating a broader reputational deterrent that reaches foreign directors of all origins, though the limited sample warrants caution.

Read together, the two markets are consistent with a single mechanism operating at different intensities of legal exposure: under intent-to-treat exposure the response is selective, falling on U.S.-connected directors while sparing other foreign directors, whereas explicit sanctions produce a broader decline in foreign directors of all origins together with measurable capital reallocation. Because the indiscriminate pole rests substantially on the eleven explicitly sanctioned firms and on the capital-base evidence, we treat it as suggestive and develop it, alongside the full international-ownership results and the cross-market synthesis, in Appendix A.2.

6 Does the Loss of U.S.-Connected Monitors Matter?

The preceding sections establish that the sanctions regime removed a specific kind of director from exposed Chinese boards, namely internationally credentialed monitors who were disproportionately independent. Whether that removal carries a governance cost is the question that gives the board-composition channel its economic stakes, because the departing directors are precisely the type that prior work associates with stronger monitoring and with constraints on controlling-shareholder expropriation in Chinese firms (Zhu et al., 2016; Gong et al., 2021; Jiang et al., 2010). To test that consequence directly, we take stock-price crash risk as our primary governance outcome. Crash risk is the most widely used market-based proxy for the consequences

[6] All Hong Kong director-exit results are robust to controlling for log market capitalization (22 of 22 specifications unchanged). We omit this control from the baseline to avoid observation loss (6.4% missing) and a potential bad-control concern, as market valuations may themselves respond to the shock.

of weak board oversight: a large literature holds that when monitoring is weak, managers can hoard adverse information until it accumulates and is released in infrequent large drops, so that a return distribution more prone to sudden crashes is the price-based signature of the opacity that weakened oversight permits. The measure is well suited to our setting. Because it is computed from each firm's own weekly returns, it is available for the full panel rather than only for the firms and years in which a discrete governance event is observed, which gives it far broader coverage than an event-based test. It also aggregates the consequences of monitoring quality into a single outcome without requiring us to pre-specify which governance margin the departing directors disciplined, an advantage where the monitoring functions at issue are diffuse. In Chinese firms, moreover, crash risk has been tied directly to board independence and to directors with overseas experience (Cao et al., 2019), the attributes the sanctions shock stripped from exposed boards. A monitoring-driven deterioration of the kind we hypothesize should therefore register in this measure if it registers anywhere. We construct the negative coefficient of skewness (NCSKEW) and the down-to-up volatility ratio (DUVOL), the two conventional crash-risk measures, for which higher values denote a return distribution more prone to sudden large declines, and regress each on the treated-by-post interaction with firm and year fixed effects, the controls of Section 3, and firm-clustered inference over 2019–2024.

We find no governance cost on this measure, and the confidence bounds are informative. As Table 12 reports, the treated-by-post coefficient is -0.016 for crash-risk skewness ($p = 0.56$) and -0.018 for down-to-up volatility ($p = 0.29$), each statistically indistinguishable from zero at conventional thresholds. The 95% confidence interval excludes effects beyond roughly one-tenth of a standard deviation in either direction, and the minimum detectable effect at 80% power is about 0.10 standard deviations, so a governance-relevant deterioration would have been detected had it occurred. The null is unmoved by pooled estimation, a lagged dependent variable, two-way clustering, the broad-market benchmark that adds the ChiNext and STAR boards, a thirty-week minimum, and dose-response specifications that scale treatment by baseline U.S.-nexus exposure or by the realized post-shock loss of U.S.-connected directors; the year-by-year event study is likewise flat, with no post-2021 break.

Table 12: Crash Risk and the Loss of U.S.-Nexus Directors

	Crash risk (neg. skewness) (1)	Crash risk (down-to-up vol.) (2)
Treated × Post (primary)	−0.016 (0.562)	−0.018 (0.288)
95% confidence interval	[−0.069, +0.037]	[−0.051, +0.015]
Robustness, Treated × Post:		
Pooled OLS	−0.016 (0.558)	−0.018 (0.277)
Lagged dependent variable	−0.017 (0.551)	−0.021 (0.246)
Two-way cluster (firm, year)	−0.016 (0.560)	−0.018 (0.347)
Broad-market benchmark	−0.014 (0.597)	−0.017 (0.330)
Minimum 30 weeks	−0.015 (0.577)	−0.018 (0.283)
Dose-response (per unit), × Post:		
Baseline U.S.-nexus exposure	+0.105 (0.732)	−0.044 (0.823)
Realized loss (count)	−0.029 (0.420)	−0.020 (0.382)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Controls	Yes	Yes
Observations (firm-years)	16,850	16,850
Firms	2,917	2,917
Min. detectable effect (80%)	≈0.10 SD	≈0.10 SD

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The dependent variables are firm-year stock-price crash-risk measures built from firm-specific weekly returns: NCSKEW, the negative coefficient of return skewness, and DUVOL, the log ratio of return volatility in below-average ("down") weeks to above-average ("up") weeks; for both, higher values indicate greater crash risk. Treated × Post is the exposure difference-in-differences interaction (Post $=$ 1 for years $\geq$ 2021); the primary specification adds firm and year fixed effects, the controls listed in Section 3, and firm-clustered standard errors over 2019–2024. The robustness rows vary that specification one change at a time; the dose-response rows replace the binary treatment with continuous baseline exposure or with the realized count of U.S.-connected directors lost, each interacted with Post, so their per-unit scale differs. The market benchmark is the Shanghai-plus-Shenzhen main-board A-share market excluding the ChiNext and STAR boards, and the broad-market row adds those boards. The minimum detectable effect is computed at 80% power against the estimated standard error. Crash-risk firm-years require at least 26 valid trading weeks.

Price-based measures such as crash risk, however, are least equipped to register the monitoring functions for which the departing directors were valued. We therefore turn to two internal-governance outcomes that bear more directly on the monitoring and minority-protection channels: the sensitivity of executive pay to firm performance and payout policy. Each is estimated as a reduced-form exposure difference-in-differences with firm and year fixed effects and firm-clustered inference over 2019–2024 on the non-financial sample, and we read the estimates as intent-to-treat consequences of baseline U.S.-nexus exposure rather than as the isolated causal effect of director loss.

Incentive alignment is the first channel. If internationally credentialed monitors disciplined the link between pay and performance, their departure should weaken it. We estimate a triple difference in which log executive compensation is interacted with firm performance, treatment, and the post indicator, so that the coefficient on the triple interaction measures the change in pay-performance sensitivity at exposed firms after the shock. Executive pay tracks performance strongly in levels—a one-unit increase in earnings per share raises top-three executive compensation by 0.139 log points (significant at the 1% level)—but the treated-by-post slope shift is economically negligible and statistically indistinguishable from zero (-0.017, $p = 0.42$; Table 13). The null is unchanged when performance is standardized within industry-year cells, when a quadratic performance term is added, under industry-by-year fixed effects, with the all-personnel compensation caliber, and over the extended 2018–2024 window. Pay-performance sensitivity does fall market-wide after 2021 (the performance-by-post coefficient is -0.094, significant at the 1% level), but this common decline falls equally on treated and control firms and is unrelated to de-internationalization.

Table 13: Pay-Performance Sensitivity and the Loss of U.S.-Nexus Directors

	(1)	(2)	(3)	(4)
Dep. var.:	ln Top-3 pay	ln Top-3 pay	ln Top-3 pay	ln Total pay
Performance:	EPS	ROA	Stock return	EPS
Performance (base slope)	0.139***	0.958***	0.094***	0.127***
	(0.000)	(0.000)	(0.000)	(0.000)
Perf. × Treated × Post	-0.017	-0.075	-0.025	-0.028
	(0.416)	(0.733)	(0.480)	(0.204)
Perf. × Treated	0.016	-0.143	0.006	0.023
	(0.483)	(0.449)	(0.840)	(0.336)
Perf. × Post	-0.094***	-0.885***	-0.056***	-0.083***
	(0.000)	(0.000)	(0.004)	(0.000)
Observations	16,591	16,593	16,216	16,591
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variable is log executive compensation (top-three senior managers in columns 1–3; all senior managers in column 4). Each column is a triple difference of pay on the indicated performance measure (basic EPS, ROA, or annual stock return), interacted with the baseline treatment indicator and the post indicator. The row of interest, Performance × Treated × Post, is the change in pay-performance sensitivity at exposed firms after 2021; the base slope reports the cross-sectional pay-performance gradient. All columns include firm and year fixed effects and the controls of Section 3 (firm size and leverage; profitability is excluded as it is the channel). Sample: non-financial firms, 2019–2024.

Payout policy is the second channel. Under the outcome model of dividends, weaker minority-shareholder protection predicts lower payouts, as controlling shareholders retain or divert cash rather than distribute it. We find no such movement. The treated-by-post effect is null for dividends scaled by assets ($+0.0004$, $p = 0.51$), for the probability of paying a dividend ($+0.011$, $p = 0.40$), for the payout ratio among profitable firm-years (-0.001, $p = 0.92$), and for post-tax dividends per share (Table 14); a continuous dose-response in baseline exposure is likewise null, and the result is unchanged under industry-by-year fixed effects and a conditional specification that adds the profitability control. The estimates are informative rather than merely imprecise, since the minimum detectable effect on the payout ratio is about four percentage points against a sample mean near thirty percent, so a payout contraction of governance-relevant magnitude would have been detected had it occurred.

Table 14: Dividend Policy and the Loss of U.S.-Nexus Directors

	(1) Dividends/assets	(2) Pays dividend	(3) Payout ratio	(4) DPS (post-tax)
Panel A. Firm and year fixed effects (unconditional)				
Treated $\times$ Post	0.000	0.011	-0.001	-0.007
	(0.514)	(0.395)	(0.918)	(0.389)
Panel B. Adds ROA control (conditional payout policy)				
Treated $\times$ Post	0.000	0.001	0.002	-0.014*
	(0.781)	(0.916)	(0.874)	(0.092)
Observations	16,634	16,634	13,288	16,634
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variables are total cash dividends scaled by assets, an indicator for paying any cash dividend, the payout ratio (cash dividends over parent net income, defined for profitable firm-years), and post-tax dividends per share (zero-filled for non-payers). Treated $\times$ Post is the exposure difference-in-differences interaction (Post $= 1$ for years $\geq$ 2021). Panel A is the unconditional specification with firm and year fixed effects; Panel B adds the profitability (ROA) control and is read as conditional payout policy. Sample: non-financial firms, 2019–2024.

Two further consequence tests, reported in the appendix for completeness, point in the same direction. The first asks how the market priced the departures themselves: in an event study of U.S.-connected director departures, cumulative abnormal returns around the departure date are economically small and statistically indistinguishable from zero across a battery of parametric and nonparametric tests, so the market registered no robust reaction to the loss of these directors (Appendix A.2, Table A4). The second is related-party transactions, the canonical tunneling channel through which controlling shareholders expropriate minority investors and which independent monitors are thought to constrain in Chinese firms (Jiang et al., 2010; Gong et al., 2021). Because the related-party-transaction data are subject to data-quality limitations that make it the noisiest of our outcomes, we treat this test as secondary; subject to that caveat, the treated-by-post effect is statistically indistinguishable from zero across the log value, asset-scaled, and count measures (Appendix A.2, Table A5).

Read together, these channels indicate that exposed firms shed internationally credentialed monitors without a detectable near-term governance cost across the price-based and internal-governance measures we observe. The most natural reading ties the null to the one-for-one domestic-professional substitution documented in Section 4: because vacated seats were refilled and board size and independence preserved, no governance vacuum opened, and the firm's opacity profile need not move. The credential comparison in Section 4 sharpens this reading, since the domestic professionals who filled the vacated seats matched the departing directors on domestic professional standing (Table 9), so that the monitoring-relevant competence a price-based proxy would register was preserved, whereas the graduate-level and international human capital that did leave is of a kind whose governance value crash risk and the market's reaction at departure are least equipped to detect. We frame this pattern as bounded resilience, since the available price-based measures cannot speak to longer-horizon or harder-to-observe costs, and we claim no causal inference here regarding the effect of board composition on real or financial outcomes. The director exit is plausibly identified by the external sanctions shock, whereas the link from board composition to crash risk or to the market's reaction at departure is subject to many confounding influences, among them contemporaneous firm performance, disclosure choices, and broader market conditions; the consequence design is in any case more demanding of the data and less sharply identified than the within-board exit hazard that carries the paper. We therefore read these results as suggestive evidence on the consequence question, corroborating the resilience-and-substitution account, with no claim that they identify the effect of board composition on firm outcomes. The internal-governance tests speak to incentive alignment and payout in the main text and to related-party tunneling in the appendix; the channels that remain least visible to prices—enforcement actions, financial restatements, and tunneling through abnormal other-receivables—require enforcement and balance-sheet data still being assembled, which we leave as future evidence (Appendix A.2).

7 Conclusion: Discussions and Limitations

With the cross-market evidence in place, this Article sets out the qualifications that bound its interpretation, taking in sequence the threats to identification, the caveats that attach to our measures, and the extensions that would tighten the design. Identification is carried by the within-firm and within-board comparisons of Section 3; the quasi-exogenous timing of the U.S. sanctions—the Entity List expansions, Executive Order 14032, and the U.S. Persons Rule—motivates the post-period indicator and the estimation window, while the causal claim rests on the within-unit comparisons. In our preferred 2019–2024 specification the single available pre-treatment coefficient (2020, base year 2019) shows no detectable divergence ($+0.0003$, not significant at the 10% level), although, as Section 3 explains, the design derives its identifying leverage from the within-firm and within-board comparisons. The extended 2018–2024 specification returns a 2018 coefficient that is negative and significant at the 1% level (-0.0075), a pattern we trace to 140 "early-treated" firms whose U.S.-nexus departures coincided with the Section 301 trade-war tariffs of July 2018; once these firms are excluded the 2018 anomaly disappears and the coefficient flips to $+0.003$. We therefore adopt the 2019–2024 window as our baseline. Were the sanction dates to arrive at genuinely different times across firms, a heterogeneity-robust estimator in the manner of Callaway and Sant'Anna (2021) would be the appropriate response; the explicit sanctions data are in any event too thin to support such standalone analysis—42 firms, 26 of them in the panel, only 7 overlapping with the treatment group—a scarcity that itself vindicates our intent-to-treat approach.

Because our treatment bundles several legal instruments together with other events of the 2020–2021 period—the pandemic, the broader trade conflict, and, in Hong Kong, the National Security Law—a natural concern is that one of these contemporaneous shocks produced the exit. The within-board design speaks to this concern as squarely as it does to mean reversion. Any shock that operates at the level of the firm, depressing all board service or all foreign board service at an exposed firm, is absorbed by the firm $\times$ year fixed effects and cannot generate the estimated gap. To survive the within-board comparison, a confounder would have to drive out a firm's U.S.-connected directors differentially relative to their own non-U.S. colleagues, and to do so on the timing of the 2021 sanctions. The pandemic, the trade war, and the National Security Law carry no such U.S.-specific, within-board signature, whereas the U.S. Persons Rule carries one by construction. The within-firm placebo reinforces this reading, since at mainland treated firms only U.S.-connected directors leave while non-U.S. foreign directors are unaffected (Table 10), which isolates the U.S.-specific component of the decoupling bundle from the concurrent market-wide de-internationalization.

The replacement analysis is descriptive, in that it characterizes who fills vacated seats among matched exit–entry pairs while leaving the causal effect of director loss on replacement type unestimated. The domestic-technocrat result ($+0.011$, significant at the 5% level) is sensitive to how we define the CCP keyword set, as a strict three-keyword definition weakens it to a coefficient that is not significant at the 10% level, whereas a broad ten-keyword definition flips the sign (-0.006, not significant at the 10% level). The sensitivity follows from the construction of the measure: because the domestic-technocrat indicator is a residual category, the boundary between party-affiliated and party-adjacent careers is inherently ambiguous. Because the replacement-pair matching uses annual data, exact exit–entry pairing is only approximate. The board-size correction nonetheless stands. Using CSMAR's pre-computed board-size measure, the coefficient is null ($+0.009$, not significant at the 10% level), which confirms complete one-for-one replacement and reveals the earlier -0.32-seat shrinkage to have been an artifact of a director-panel count that included supervisory and executive positions. Note that the replacement result also weakens to a coefficient that is not significant at the 10% level when we exclude the 140 early-treated firms ($+0.004$), consistent with those firms driving part of the effect.

The mainland arm shows no institutional-ownership response at treated A-share firms across three independent sources, a null that is consistent with a capital-base channel operating only where baseline foreign exposure is substantial; the full mainland ownership nulls appear alongside the Hong Kong capital-base evidence in Appendix A.2. The Hong Kong surname classifier is a noisy proxy, in that it cannot distinguish a Hong Kong permanent resident with a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. Our multi-dialect approach reduces this error without eliminating it, and the independent CSMAR-sourced nationality classifier (70.5% coverage of Hong Kong director-years, with a 25-firm overlap with the surname group) supplies the triangulating cross-check on which we rely throughout. Because the explicit-sanctions sample comprises only 11 firms, all of them large SOEs, the strong ownership results may not generalize to smaller or privately owned firms, and the Stock Connect sign reversal between sanctioned and intent-to-treat firms further complicates the ownership interpretation. Several of the cross-market statements we retain lean on the eleven-firm coefficient, and we accordingly report them tentatively.

Several extensions would strengthen the analysis. With the within-board exit-hazard design in Section 3 now differencing out firm-level mean reversion directly, an explicit director-tenure control is the main remaining refinement to the identification battery. Firm-level registry data on Qualified Foreign Institutional Investor (QFII) and Renminbi QFII (RQFII) holdings, in place of the top-10 approximations, would sharpen the capital-base test. A further avenue would ask whether the U.S.-connected directors left around the time their own firm was named for sanction, which would tie the exits more closely still to firm-specific legal exposure if departures followed each listing date. This reading is unavailable in practice. Few of the firms we study were named individually, and fewer still had U.S.-connected directors before the shock, so the firms that could in principle reveal firm-specific timing are too few to support it. The named firms came under the binding investment prohibition at essentially the same moment, in any case, so their listing dates carry little of the spread in timing that such a design would require. That scarcity itself reinforces the broader, exposure-based reading we adopt throughout, which does not turn on resolving any single firm's sanction date.

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A Appendix

A.1 Measurement Validation

Our empirical exercise requires an attribute that standard corporate-governance data do not report directly, namely whether an individual director holds a meaningful connection to the United States. We construct this attribute through a staged measurement procedure whose primary layer rests on auditable structured data and is supplemented, for the cases that structured fields cannot resolve, by a classification layer.

Our primary measure classifies a director as U.S.-nexus whenever any of three conditions holds in CSMAR structured records: (i) U.S. nationality or permanent residency, (ii) at least five years of U.S. work experience, or (iii) a U.S. professional license. We apply this definition to the resume, education, professional-title, nationality, and licensure fields of the CSMAR Corporate Governance module. Because this structured measure is fully auditable and does not depend on any model-based inference, we treat it as the regressor underlying all of our headline estimates.

For directors whose U.S. connection we cannot verify from structured fields, we supplement the structured measure with biographical text retrieved from public web sources (i.e., Baidu search snippets), which a large-language model then classifies into a nationality / U.S.-nexus label. This procedure yields the market-specific classified files that we use in our robustness analysis. We treat the classification layer as supplemental, since it defines the expanded treatment definitions (v7 and v8 in Appendix A.3) while leaving the primary measure unchanged. As Appendix A.3 documents, the headline exit estimate attenuates only mildly once we add these classified directors (from -0.020 to -0.018), which confirms that the result survives the addition of model-based labels. We therefore present the auditable three-prong construct as our primary measurement approach and the classification layer as a robustness extension.[7]

In the Hong Kong market, where structured nationality coverage is thinner, we infer director origin from a multi-dialect Chinese surname classifier (i.e., a classifier spanning Mandarin, Cantonese, Hokkien, and Hakka) that flags non-Chinese names. We validate this surname measure against an independent CSMAR-sourced nationality classifier, which covers 70.5% of Hong Kong director-years. The two instruments appear to capture largely distinct director populations, since the surname-based and nationality-based treatment groups overlap on only 25 firms and neither treatment predicts the other's dependent variable (cross-DV checks, not significant at the 10% level). We read this pattern as triangulation, because the classifiers measure distinct populations and thereby serve as mutual cross-checks. The surname classifier is admittedly noisy, since it cannot distinguish a Hong Kong permanent resident bearing a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. We therefore

[7] The classification layer queries the DeepSeek `deepseek-chat` model on retrieved biographical text, with the supplementary three-prong reclassification run at a sampling temperature of 0.1, and the classification prompts are archived for reproducibility. Because this layer enters only the expanded treatment definitions (v7 and v8) and leaves the headline estimates essentially unchanged, we treat its full reproducibility and accuracy validation—fixed-temperature reruns and precision and recall against a hand-labeled audit sample—as supplemental, to be reported should a referee ask that the classifier be foregrounded.

treat the surname measure as a broad indicator of Western-aligned directors and the nationality measure as the narrower, U.S.-specific instrument; in Section 5 we exploit the gap between the two to separate the U.S.-specific channel from broader de-internationalization.

A.2 Capital-Base Response and the Compliance-Chill Threshold

This appendix collects the cross-market capital-base evidence and the full selective-versus-indiscriminate synthesis demoted from the main text, together with the corroborating consequence proxies that triangulate Section 6. We present this material here because it is auxiliary to the within-board exit design that carries the main identification claim.

A.2.1 International Ownership: Explicitly Sanctioned Firms

Table A1: International-Ownership Decline: Explicitly Sanctioned Hong Kong Firms

	(1) International	(2) International	(3) Mainland	(4) Stock Connect
DV: Ownership share (%)				
Sanctioned × Post	-11.05^{***}	-10.36^{***}	$+3.04^{**}$	$+11.96^{***}$
	(<0.001)	(<0.001)	(0.020)	(0.004)
Controls	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	79,495	71,510	71,509	20,120
Firms	2,398	2,307	2,307	739
R^2 (within)	-0.0003	0.0041	0.0005	0.1038

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses the Hong Kong quarterly ownership panel for explicitly sanctioned firms and reports ownership-share outcomes with firm and quarter fixed effects.

Table A1 tests whether explicitly sanctioned Hong Kong-listed firms experienced a decline in international institutional ownership. Treatment is binary, comprising the 11 firms designated on U.S. sanctions lists. The quarterly panel comprises 89,621 firm-quarter observations (2,398 firms) in the raw panel, with ownership breakdowns sourced from Wind Terminal, while the estimation samples are smaller after controls and coverage restrictions (Table A1). Sanctioned firms experienced a 10.4 pp decline in international ownership (significant at the 1% level), accompanied by offsetting increases in mainland Chinese ownership ($+3.0$ pp, significant at the 5% level) and Stock Connect holdings ($+12.0$ pp, significant at the 1% level). The reallocation appears to operate through the northbound/southbound Connect mechanism, with direct mainland purchases playing little role. In the broader intent-to-treat sample, by contrast, ownership effects are uniformly null for international and Chinese ownership (Table A2; all not significant at the 10% level), leaving only a negative Stock Connect effect. Measurable financial-capital reallocation thus appears only among firms subject to legally binding investment prohibitions, although the intent-to-treat nulls partly reflect lower baseline foreign exposure and therefore provide limited evidence on investor non-response.

A.2.2 International Ownership: Intent-to-Treat

Table A2 extends the ownership analysis to the broader intent-to-treat sample, using the Western-surname and foreign-dialect treatment definitions. The results are uniformly null for international and Chinese ownership, since none of the four coefficients approaches significance (not significant at the 10% level). The only significant result is a negative Stock Connect effect (-2.47, significant at the 1% level for Western treatment; -3.12, significant at the 1% level for foreign-dialect). This sign reversal relative to the explicitly sanctioned firms (Table A1) likely reflects compositional differences. Explicitly sanctioned firms are predominantly large SOEs that attract Stock Connect inflows as mainland investors substitute for departing international holders, whereas the broader intent-to-treat sample includes smaller firms in which foreign-director presence signals international orientation without triggering forced ownership reallocation. Ownership concentration (top-10 shareholder share) is null across all specifications (not significant at the 10% level). These nulls constitute the financial-capital half of the threshold result: measurable capital reallocation appears to require the binding prohibition of explicit sanctions, while mere geopolitical exposure yields null ownership effects and limited evidence, because baseline foreign exposure is low.

A.2.3 Mainland Ownership: No Measurable Response

The mainland arm finds no institutional-ownership shift at treated A-share firms. Three data sources independently produce null aggregate results: a top-10 shareholder origin decomposition, aggregate institutional ownership (CSMAR; -0.53pp, not significant at the 10% level), and QFII-specific foreign ownership (-0.017pp, not significant at the 10% level). The QFII data strongly support the structural-exposure interpretation, since sanctioned A-share firms held only 0.03% QFII pre-shock, leaving essentially no foreign capital to lose. Two qualifications temper this categorical null. A dose-response test among the most-exposed firms yields a significant negative effect (-10.73, significant at the 5% level), which suggests a weak, transitory channel in the subset with the highest pre-shock U.S.-nexus exposure, while the top-10 analysis shows a small increase in Hong Kong nominee/HKSCC clearing-account ownership ($+0.16$, significant at the 5% level), likely reflecting Stock Connect reallocation. These signals do not survive as robust aggregate findings. The mainland null is consistent with a capital-base substitution channel that operates only where baseline foreign exposure is substantial, which is why a measurable effect appears only among explicitly sanctioned Hong Kong-listed firms, where baseline international ownership stood at approximately 26%.

Table A2: International Ownership: Intent-to-Treat (Hong Kong Quarterly Panel)

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: International ownership (%)		DV: Mainland ownership (%)		DV: Stock Connect ownership (%)	
	Western	Foreign	Western	Foreign	Western	Foreign
Treated × Post	$+2.42$	$+0.97$	$+1.36$	-0.51	-2.47^{***}	-3.12^{***}
	(0.226)	(0.215)	(0.227)	(0.558)	(<0.001)	(<0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Observations	71,510	71,510	71,509	71,509	20,120	20,120
Firms	2,307	2,307	2,307	2,307	739	739
R^2 (within)	0.0078	0.0113	0.0005	0.0004	0.0601	0.0170

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports broader intent-to-treat ownership estimates for Western-surname and foreign-dialect treatment definitions in the Hong Kong quarterly panel.

A.2.4 The Compliance-Chill Threshold

The cross-market evidence is consistent with a single mechanism operating at different intensities of legal exposure. Under compliance chill (i.e., intent-to-treat exposure with no binding prohibition), the response is selective: only U.S.-connected directors leave, whereas non-U.S. foreign directors are unaffected on the mainland and are recruited in Hong Kong. Under legal compulsion through explicit sanctions, the response becomes indiscriminate, as foreign directors of all origins decline and international capital reallocates. Table A3 lays the three regimes side by side.

Table A3: Selective Compliance Chill versus Indiscriminate Sanction: Cross-Market Board and Ownership Response

	Mainland (ITT)	HK (ITT)	HK (Sanctioned)
U.S./Western director exit	-0.020^{***}	-0.013^{**} (U.S.-nat.); -0.025^{***} (surname)	-0.014 (n.s., n=11)
Non-U.S. foreign response	$+0.001$ (n.s.)	$+0.010^{**}$	-0.029^{*}
Replacement type	Domestic technocrat	Non-U.S. international	Not directly measured (foreign share declines)
Intl. ownership response	Null (QFII $\approx$0.03% baseline)	Null; Stock Connect -2.47^{***}	Intl -10.4^{***}; CN $+3.0^{**}$; Connect $+12.0^{***}$
Selectivity	Selective	Selective (substitution)	Indiscriminate

Note: Cell entries are difference-in-differences coefficients on director-share or ownership-share outcomes; parenthetical notes give significance or classifier. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Treated firms: mainland 1,109; Hong Kong ITT 137 (U.S.-nationality); Hong Kong sanctioned 11. The sanctioned U.S.-nationality cell is imprecise ($n = 11$); the indiscriminate reading rests on the significant non-U.S. (-0.029^{*}) and international-ownership (-10.4pp) cells rather than that point estimate. Sources: Tables 3, 7, 8, 11, A1, 10, and A2.

The contrast indicates that human capital and financial capital respond at different thresholds of geopolitical pressure: director exit appears across the full intent-to-treat sample, whereas measurable ownership reallocation appears only where holding a specific security became legally prohibited. We read this as suggestive evidence, since the ownership nulls partly reflect negligible baseline foreign exposure—A-share sanctioned firms held only about 0.03% QFII pre-shock, which limits what they can reveal about investor response—and "compliance chill" is our interpretation inferred from the director pattern.

A.2.5 Corroborating Consequence Proxies

This appendix reports in full the two consequence tests that Section 6 alludes to but does not tabulate in the main text: the market's reaction to U.S.-connected director departures, and related-party transactions as a tunneling proxy. Both tests return bounded nulls consistent with the crash-risk, pay-performance, and payout results reported there.

The market's reaction at departure. We first ask how the market priced the departures themselves. The events are U.S.-connected director departures matched to the director panel, with informative departures separated from scheduled term-expiries. Abnormal returns come from a one-factor market model estimated on log returns over the $[-250, -30]$ window of at least 120 trading days, applying a trade-to-trade correction for the thin trading common among Chinese stocks.[8] Cumulative abnormal returns (CARs) are then aggregated over symmetric windows after collapsing multiple departures at the same firm and date. Inference draws on a battery of tests: the Patell and Boehmer-Musumeci-Poulsen standardized-residual tests, the generalized sign test benchmarked on the estimation window, the Wilcoxon signed-rank test, and the cross-sectional Student-t, supplemented by Kolari-Pynnonen cross-correlation-adjusted and GRANK-t variants. Were the loss of a U.S.-connected monitor read as value-destroying, one would expect a negative and significant cumulative abnormal return at the departure date; instead the cumulative abnormal returns are economically small and statistically indistinguishable from zero across this battery (Table A4), so the market registered no robust reaction to the loss of U.S.-connected directors. Panel A reports cumulative abnormal returns by departure group across event windows under the primary market model, where the magnitudes are small and the scattered significance in the scheduled and pooled groups tracks term-expiry calendar clustering with no U.S.-specific reaction. Panel B shows that the only positive flicker, the informative independent departures over $[-2, 2]$, is model-sensitive. The cumulative abnormal return is stable in magnitude across market benchmarks and model specifications, yet it attains significance only under the three-factor model and only on the standardized-residual tests, failing the cross-sectional Student-t and the nonparametric tests throughout.

[8] The thin-trading correction follows Maynes and Rumsey (1993).

Table A4: Director-Departure Cumulative Abnormal Returns

Panel A. CAARs by departure group and event window (market model, main-board A-share benchmark)

	$[0, 0]$	$[-1, 1]$	$[-2, 2]$	$[-5, 5]$	$[-10, 10]$
U.S.-nexus informative, post (N=225)	$+0.01$	-0.11	$+0.09$	-0.50	-0.65
of which independent (N=56)	$+0.13$	$+0.44$	$+1.54^{*}$	-0.15	$+0.71$
U.S.-nexus all departures, post (N=595)	-0.07	$+0.06$	$+0.38^{*}$	$+0.17$	-0.40
U.S.-nexus scheduled, post (N=372)	-0.11	$+0.15$	$+0.54^{*}$	$+0.59^{*}$	-0.32
U.S.-nexus informative, pre (N=132)	$+0.12$	$+0.43$	$+0.27$	-0.39	-0.69

Panel B. Informative independent departures (post; $N = 56$): the $[-2, 2]$ reaction across models and benchmarks

	CAAR%	Student-t	Patell	BMP	Sign	Wilcoxon
Market model, main-board A-shares	$+1.54$	1.37	1.67*	0.98	0.94	1.27
Market model, broad A-shares	$+1.52$	1.36	1.60	0.95	1.00	1.26
Market model, ST/halt-excluded	$+1.28$	1.08	1.40	0.80	0.79	1.03
Three-factor, main-board A-shares	$+1.77$	1.79*	2.78***	2.09**	1.67*	1.00
Three-factor, broad A-shares	$+1.79$	1.81*	2.65***	1.93*	1.91*	1.15

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Cumulative abnormal returns (CAARs) are in percent, averaged across departure events after collapsing multiple departures at the same firm and date; event day 0 is the director's recorded departure date, as CSMAR reports no separate announcement date. Abnormal returns are from a one-factor market model (Panel A and the market-model rows of Panel B) or a Fama-French three-factor model (the three-factor rows), estimated on log returns over $[-250, -30]$ (minimum 120 trading days) with a trade-to-trade thin-trading correction (Section 6). "Main-board A-shares" is the Shanghai-plus-Shenzhen A-share market excluding the ChiNext and STAR boards, and "broad A-shares" adds those boards. In Panel A, significance markers are from the Patell test, and the "of which independent" row is the independent-director subset of the informative-departure group. Panel B reports test *statistics*: Student-t is the cross-sectional t; Patell and BMP are the standardized-residual tests; Sign is the generalized sign test benchmarked on the estimation window; and Wilcoxon is the signed-rank statistic. For the primary market-model focal cell the Kolari-Pynnonen-adjusted Patell statistic (0.55) and the GRANK-t statistic (1.24) are likewise insignificant, and a cross-sectional regression of the $[-2, 2]$ abnormal return on the U.S.-nexus-by-post interaction yields -0.50 percentage points ($p = 0.50$). The benchmark adding only ChiNext (excluding STAR) lies between the two reported benchmarks and yields the same pattern.

Related-party transactions. The second test is related-party transactions, the canonical tunneling channel through which controlling shareholders expropriate minority investors and which independent monitors are thought to constrain in Chinese firms (Jiang et al., 2010; Gong et al., 2021). Were the loss of U.S.-connected monitors to loosen that constraint, related-party activity should rise. We report this test as secondary rather than in the main text, because the related-party-transaction data are subject to data-quality limitations that make it the noisiest of our outcome measures. Subject to that caveat, the evidence aligns with the main-text channels. The treated-by-post effect is statistically indistinguishable from zero for the log value of related-party transactions ($+0.16$, $p = 0.38$), for related-party transactions scaled by assets (-0.003, $p = 0.64$), and for the count of transactions (-0.76, $p = 0.35$). These nulls survive the addition of industry-by-year fixed effects (Table A5). The asset-scaled measure is the most tightly estimated, with a minimum detectable effect near 1.9 percentage points of assets, whereas the log and count margins are less precise and cannot exclude a moderate response.

Table A5: Related-Party Transactions and the Loss of U.S.-Nexus Directors

	(1) ln RPT	(2) RPT/assets	(3) RPT count
Panel A. Firm and year fixed effects			
Treated × Post	0.156	-0.003	-0.763
	(0.379)	(0.640)	(0.346)
Panel B. Firm and industry × year fixed effects			
Treated × Post	0.181	-0.002	-0.968
	(0.319)	(0.730)	(0.223)
Observations	16,634	16,634	16,634

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values; standard errors are clustered by firm. The dependent variables are the log value of related-party transactions, related-party transactions scaled by assets, and the count of related-party transactions. Treated × Post is the exposure difference-in-differences interaction. Panel A uses firm and year fixed effects; Panel B replaces year with industry × year fixed effects. Sample: non-financial firms, 2019–2024.

The remaining proxies that bear on channels less visible to prices—the incidence of enforcement actions and financial restatements, and abnormal other-receivables as a tunneling measure—require enforcement and balance-sheet data still being assembled, and we leave them as future evidence. We therefore treat them as the next natural step for evaluating the consequence channel beyond return-based measures.

A.3 Robustness and Pre-Trend Diagnostics

A.3.1 Treatment-Definition Robustness

Table A6: Treatment-Definition Robustness: Main Results and Placebo Tests

	v5 (CSMAR) N_T=1,109	v7 (+scraper nat.) N_T=1,164	v8 (+scraper 3-prong) N_T=1,298
Panel A: Main Results			
Director exit (share)	-0.0200^{***}	-0.0190^{***}	-0.0180^{***}
	(<0.001)	(<0.001)	(<0.001)
Director exit (count)	-0.3692^{***}	-0.3487^{***}	-0.3375^{***}
	(<0.001)	(<0.001)	(<0.001)
Replacement (domestic technocrat)	$+0.0108^{**}$	$+0.0103^{**}$	$+0.0100^{**}$
	(0.024)	(0.024)	(0.027)
Panel B: Placebo Tests			
Placebo T1 (same firms)	+0.0007	+0.0010	+0.0011
	(0.484)	(0.319)	(0.279)
Placebo T2 (unexposed)	-0.0161^{***}	-0.0165^{***}	-0.0169^{***}
	(<0.001)	(<0.001)	(<0.001)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. v5 uses the CSMAR three-prong heuristic; v7 adds Baidu-scraper-classified U.S. nationals; v8 adds scraper work experience and licenses. Placebo T1 uses non-U.S. foreign share at treated firms, and T2 uses non-U.S.-foreign-treated firms with no U.S. exposure. All specifications include firm and year fixed effects, controls, and firm-clustered standard errors.

Table A6 reports our results under three treatment definitions. The primary specification (v5) relies on CSMAR structured data to classify U.S.-nexus directors via the three-prong heuristic, identifying 1,109 treated firms. The expanded definitions add directors recovered from Baidu search snippets, so that v7 incorporates scraper-classified U.S. nationals (+55 firms), while v8 further adds directors with U.S. work experience or U.S. licenses (+134 firms).

We adopt v5 as our primary specification because the v7/v8 additions introduce measurement noise. The scraper-identified directors differ systematically from the CSMAR-identified population: they are predominantly non-independent (75–78% vs. 52%), they exhibit weaker post-shock exit rates (-25.5% vs. -41.9% at treated firms over 2020–2024), and 73–74% have zero observations in the analysis window, having served before the sanctions period. These characteristics are consistent with the scraper capturing less prominent directors whose U.S. connections are harder to verify from structured data.

Our results remain robust across all three definitions. Panel A shows that the exit estimate attenuates monotonically from -0.020^{***} (v5) to -0.018^{***} (v8), consistent with adding firms whose

U.S.-nexus directors exit at lower rates. The replacement result is stable ($+0.011$** to $+0.010$**), while Panel B confirms that the placebo tests are invariant to the treatment definition.

A.3.2 Cross-Specification Robustness Summary

Table A7 consolidates the key coefficient estimates across our retained hypotheses and extensions. Panel A reports the main results, whereas Panel B reports the extensions, including the corrected board-size result ($+0.009$, null) and the independent-director ratio (-0.086, null), together with the SOE executive-pay decline (-0.057, significant at the 5% level). Panel C reports industry $\times$ year fixed-effect specifications, under which the exit result is virtually unchanged (-0.020***). Panel D reports the most conservative clustering. Because the baseline clusters on roughly 2,941 firms, asymptotic inference is already reliable at the firm level, so we present the more demanding clustering schemes as reassuring sensitivity checks. The exit result remains highly stable under two-way firm and year clustering—itself a conservative sensitivity check, since the year dimension carries only six clusters—where the standard error increases roughly fourfold yet remains significant at the 1% level, and the replacement result survives. As a further check at the dimensions where cluster counts are small enough to threaten the asymptotic approximation, the wild cluster bootstrap confirms the exit result (significant at the 1% level) under both industry (18 clusters) and province (36 clusters) bootstraps.

Table A7: Cross-Specification Robustness Summary

	DV	$\hat{\beta}$	p	N	Sample
Panel A: Main Results					
H1	U.S.-nexus director share	-0.0200^{***}	<0.001	17,171	2019–2024
H2	Domestic technocrat	$+0.0108^{**}$	0.024	314,050	2019–2024
Panel B: Extensions					
Ext 1a (dose)	U.S.-nexus director share	-0.3692^{***}	<0.001	17,171	2019–2024
Ext 1b	Euphemistic resignation	-0.0652^{**}	0.030	54,117	2019–2024
Ext 1c (Independent)	U.S.-nexus share (independent)	-0.0477^{***}	<0.001	17,171	2019–2024
Ext 2a (Technical-background index)	Technical-background index	$+0.0058^{**}$	0.011	314,050	2019–2024
Ext 2b (Board size)	Board size	$+0.0090$	0.821	17,110	2019–2024
Ext 2b (Independent ratio)	Independent-director ratio	-0.0859	0.611	17,109	2019–2024
Ext 2c (POE indep)	Log(indep. director pay)	$+0.1596^{**}$	0.028	5,437	2019–2024
Ext 2c (TOP3, SOE)	Log(top-3 exec. pay)	-0.0574^{**}	0.022	5,467	2019–2024
Panel C: Industry $\times$ Year FE Robustness					
H1 (Ind$\times$Yr)	U.S.-nexus director share	-0.0198^{***}	<0.001	17,171	2019–2024
Panel D: Alternative Clustering (Most Conservative)					
H1 (Firm+Year)	U.S.-nexus director share	-0.0200^{***}	<0.001	17,171	2019–2024
H2 (Firm)	Domestic technocrat	$+0.0108^{**}$	0.024	314,050	2019–2024

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. All specifications include firm and year fixed effects unless noted; the replacement analysis uses replacement-pair data with the treated-U.S.-exit indicator as the treatment variable.

A.3.3 Director-Tenure Robustness

Table A8: Director Tenure and the Within-Board Exit Gap

	(1) Baseline	(2) + Tenure control
U.S.-nexus × 2018	−0.0184	−0.0044
	(0.128)	(0.706)
U.S.-nexus × 2019	+0.0246*	+0.0348***
	(0.060)	(0.006)
U.S.-nexus × 2021	+0.0171	+0.0182
	(0.197)	(0.143)
U.S.-nexus × 2022	+0.0370***	+0.0377***
	(0.006)	(0.003)
U.S.-nexus × 2023	+0.0442***	+0.0393***
	(0.002)	(0.004)
Firm × Year FE	Yes	Yes
Director-tenure control	No	Yes
Director-years	219,282	219,282
Firms (clusters)	2,048	2,048

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values from standard errors clustered by firm. The dependent variable equals one if a director leaves the board before the following year. Each coefficient is the within-board difference, between U.S.-connected directors and their non-U.S. colleagues on the same board in the same year, by calendar year relative to the 2020 base. Both columns include firm × year fixed effects, so identification rests only on differences among directors serving on the same board in the same year. Column 2 adds the director's board tenure, in years served, and its square. The two columns are estimated on the common sample of firms for which director appointment dates are available (2,048 of 2,950 firms).

A further concern is that U.S.-connected directors may leave at higher rates because they hold younger directorships, which turn over more often for reasons unrelated to the sanctions. Because U.S.-nexus directors do serve shorter tenures than their board colleagues—a pre-2021 average near three years against three-and-a-half—we remove this channel directly. Table A8 re-estimates the within-board event study with a control for director tenure and its square, on the firms for which appointment dates are available. The post-shock widening of the within-board exit gap is essentially unchanged, the 2022 and 2023 coefficients moving only from $+0.037$*** and $+0.044$*** to $+0.038$*** and $+0.039$***, so the rising gap does not appear to reflect tenure composition. The control does raise the 2019 pre-shock coefficient to $+0.035$***, a within-board divergence we attribute to the 140 early-treated firms whose U.S.-nexus departures coincided with the July 2018 Section 301 tariffs; the within-board design does not in any case rest on pre-trend parallelism. Appointment dates are available for 2,048 of the 2,950 sample firms, so the control is estimated on that common subsample.

A.3.4 Pre-Trend Diagnostics

Our preferred mainland specification (2019–2024, Post $\geq$ 2021) has a deliberately thin pre-period. The single available pre-treatment coefficient (2020 relative to the 2019 base year) is $+0.0003$ (not significant at the 10% level), showing no detectable pre-shock divergence; parallelism cannot be assessed from one pre-period and is not the design's identifying assumption, with the within-firm placebo (Table 10) supplying the operative falsification against mean reversion. The event study shows a monotonic post-shock decline from -0.009 (2021) to -0.032^{***} (2024). The extended 2018–2024 mainland event study reveals a significant 2018 coefficient (-0.0075, significant at the 1% level), which we trace to 140 "early-treated" firms with U.S.-nexus departures coincident with the Section 301 trade-war tariffs (July 2018); excluding these firms eliminates the anomaly. In Hong Kong, the surname-based event study shows significant pre-trend coefficients in 2015 (-0.030^{***}) and 2016 (-0.020^{**}), yet the relevant 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the earlier anomaly likely reflects the onset of board internationalization among Hong Kong-listed firms in the mid-2010s. The nationality-based event study displays a similar pattern, with 2017–2019 clean (all not significant at the 10% level) and a monotonic post-shock decline reaching -0.029^{***} by 2024.

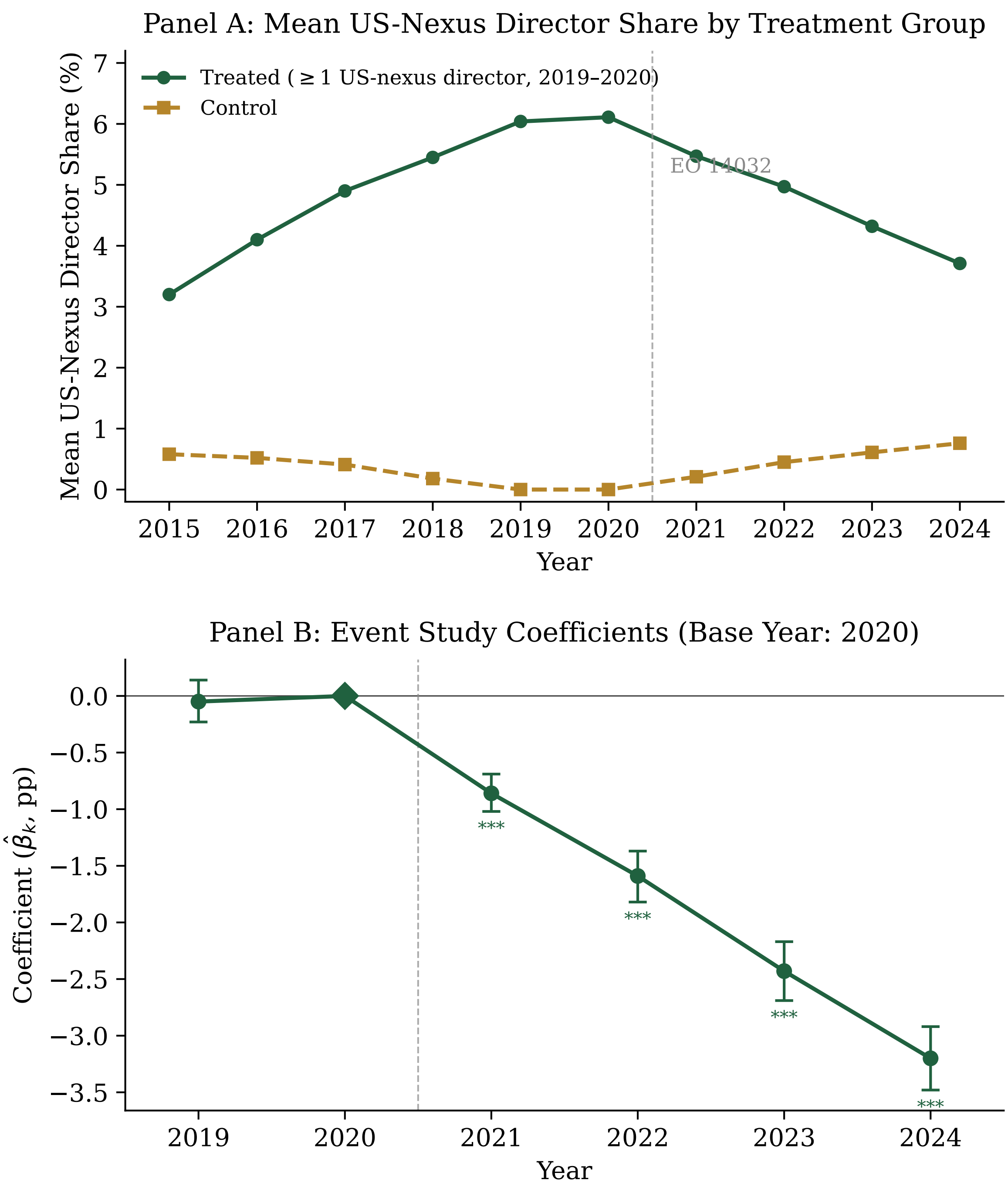

Figure A1: Event-study pre-trends, mainland A-share firms. Panel A plots the mean U.S.-nexus director share for treated (≥ 1 U.S.-nexus director in 2019–2020) versus control firms; Panel B plots the year-by-treatment event-study coefficients (base year 2020) with 95% confidence intervals. The single available pre-treatment coefficient (2019, not significant at the 10% level) shows no detectable divergence, and the monotonic post-2021 decline is consistent with a shock-driven exit; the design's causal weight rests on the within-firm and within-board comparisons set out in the text, which a one-year pre-window cannot supply. Exact coefficients appear in Table 4.

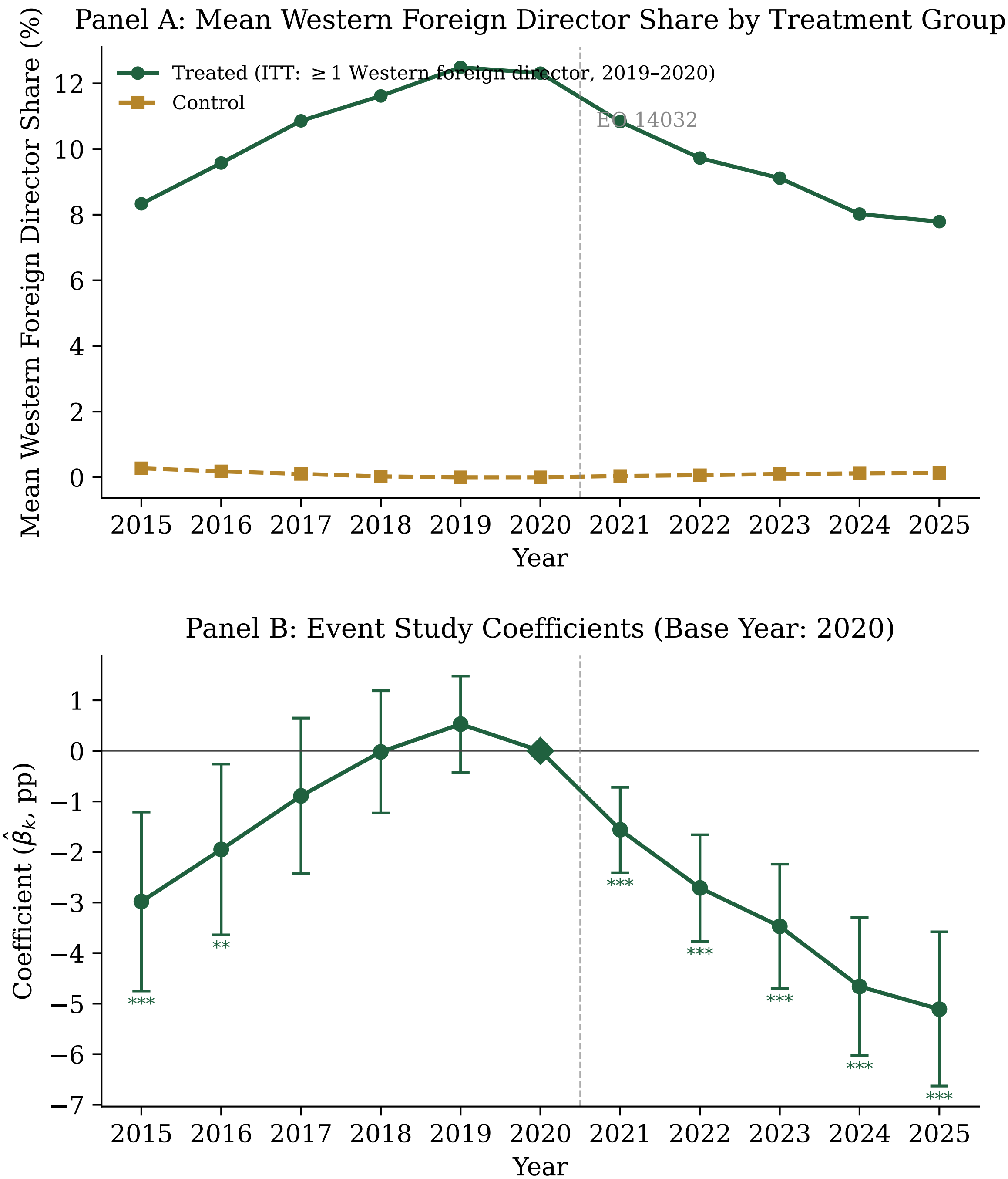

Figure A2: Event-study pre-trends, Hong Kong-listed Chinese firms. Panel A plots the mean Western-foreign director share for treated versus control firms; Panel B plots the year-by-treatment event-study coefficients (base year 2020) with 95% confidence intervals. The 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the larger 2015–2016 coefficients reflect the onset of board internationalization in the mid-2010s rather than a sanctions response. The post-2021 decline mirrors the mainland pattern.

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